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Software Requirements Specification

For

Version [01]

Students:

Supervised By:
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Revision History

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Table 1 Revision History

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1. Introduction

The Software Requirements Specification (SRS) for EDUassist provides a comprehensive overview of the academic management tool designed to assist students in organizing their educational responsibilities. This document outlines the functional and non-functional requirements essential for the design, serving as a critical reference for software engineers and stakeholders. It details the system's intended functionalities, benefits, and constraints, ensuring clarity and alignment with user needs. By following the guidelines presented in this SRS, the development team will be equipped to create an effective solution that enhances the educational experience for students.

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to delineate the software requirements for the EDUassist project. This document offers a detailed description of the project's objectives, intended functionalities, and constraints, ensuring that all stakeholders have a clear understanding of its scope and expectations. It serves as a vital reference for developers, project managers, and other stakeholders, facilitating effective communication and guiding the project's development and implementation. By providing a structured framework, this SRS aims to align all parties involved and support the successful realization of the EDUassist software product.

1.2 Scope

This subsection outlines the scope of the EDUassist software product, detailing its intended functionalities, limitations, and applications as described in this Software Requirements Specification (SRS). The primary focus is the EDUassist software, envisioned as a comprehensive academic management tool aimed at assisting students in effectively managing their educational responsibilities through a structured framework.

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Roles	Functionalities Provided
Admin:	• Create, modify, and delete user accounts for end users. • Reset passwords and manage user roles. • Monitor system performance . • Configure system settings and integrations with external tools.
End User:	• Sign in to access personalized study plans and resources. • Generate customized study plans based on academic goals and provide files . • Track academic tasks and deadlines. • Utilize an AI-assisted whiteboard for visualization and collaboration. • Generate short explanatory videos from whiteboard sessions. • Access a library of educational resources and receive personalized recommendations. • Share resources and collaborate with peers.
System (EDUassist):	• Provide secure authentication and manage sign-in/sign-out processes. • Support password recovery . • Store user profiles, study plans ,files, and progress tracking data securely. • Integrate with external calendar services for task management. • Track user interactions and generate usage statistics for monitoring. • Provide ai assistant .

Table 2 Functionalities of each actor role

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- **Functionalities:**
  - **What EDUassist Will Do:**
    - **Personalized Study Plans:** EDUassist will generate individualized study plans based on user-defined academic goals and learning preferences, helping students effectively schedule their study time and prioritize tasks.
    - **Collaborative Tools**: will be include features that facilitate collaboration among students, enabling resource sharing and effective .
    - **Integration Capabilities**: The specifications detail potential integration with existing platforms such as Google Calendar and Learning Management Systems (LMS) to enhance user productivity through seamless data synchronization.
    - **Resource Recommendations:** The proposal includes the use of artificial intelligence to provide personalized recommendations for study materials, tailored to each user's academic progress and preferences.
    - **Task Tracking:** The framework incorporates provisions for tracking academic tasks and deadlines, assisting users in maintaining organization and focus throughout their studies.
    - **AI-Assisted Whiteboard:** A key feature will be an interactive whiteboard equipped with AI assistance, helping students visualize complex concepts through drawing, annotation, and real-time collaboration. The AI component will analyze user inputs and provide contextual suggestions or explanations.
    - **Video Generation:** To enhance understanding, the system will allow for the generation of short explanatory videos based on content discussed on the whiteboard, enabling students to revisit challenging topics in a dynamic format.
  - **What EDUassist Will Not Do:**
    - **Replacement of Existing Platforms:** EDUassist is not intended to replace existing educational tools; rather, it aims to complement them by enhancing their functionalities.

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- **Grading and Assessment Functions:** The proposed system will not include grading or assessment capabilities, focusing solely on academic management and resource organization.

- **Application :**

  - **Relevant Benefits, Objectives, and Goals:**

    - **Enhanced Academic Organization:** The EDUassist framework aims to improve organizational capabilities for students by providing structured study plans and task management tools, thereby alleviating academic stress and enhancing efficiency.

    - **Facilitation of Collaboration:** This outline emphasizes the importance of collaborative learning, proposing tools that encourage teamwork and resource sharing to enrich the educational experience.

    - **Improved Resource Accessibility**: EDUassist is designed to enhance the accessibility of relevant study materials, optimizing the learning process and encouraging effective study habits.

    - **Increased Time Efficiency:** aims to minimize the time needed for students to plan and organize their studies, incorporating features such as automated reminders and integrated calendar functionalities.

    - **Enhanced User Engagement:** By proposing an intuitive interface and tailored resources, EDUassist seeks to foster greater student engagement and motivation in their academic pursuits.

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1.3 Definitions, Acronyms, and Abbreviations

Terminologies are listed in the table below:

Terminology	Definition
EDUassist	A platform designed to assist students in managing studies and collaborating using AI tools.
Study Plan	A personalized schedule tailored to students' syllabus and preferences for tracking progress.
AI Recommendations	Suggestions generated by the system to help users access relevant learning resources.
Whiteboard Collaboration	A shared virtual workspace enabling students to interact and annotate in real time.
Admin	A system user responsible for managing user accounts, monitoring progress, and sending alerts.
Student	The primary user of EDUassist, utilizing features like study plans and collaborative tools.
Notifications	Alerts sent by the system to inform users of deadlines, updates, or recommendations.
External Tools	Third-party apps like Google Calendar or Zoom integrated into the platform.

Table 3 Terminologies

Acronyms are listed in the table below:

Acronym	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
WCAG	Web Content Accessibility Guidelines
GDPR	General Data Protection Regulation
OTP	One-Time Password
LMS	Learning Management System
UML	Unified Modeling Language
DFD	Data Flow Diagram
STD	State-Transition Diagram
SRS	Software Requirements Specification

Table 4 Acronym

Abbreviations are listed in the table below:

Abbreviation	Meaning
Dashboard	The main interface for users to interact with the system.
Task Tracker	A feature for monitoring tasks, deadlines, and progress.
Collab	Short form for collaboration in the whiteboard feature.
RecSys	Abbreviation for Recommendation System.
Processes	Actions within the DFD, such as generating study plans or managing user profiles.
Entities	External actors in the DFD, such as students, admins, or third-party tools.

Table 5 Abbreviations

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1.5 Overview

This section provides a summary of the Software Requirements Specification (SRS) structure and content. The SRS is organized to offer a comprehensive outline of the EDUassist system, detailing its objectives, functionalities, and design constraints. It begins with an introduction to the system’s purpose and scope, ensuring that all stakeholders have a clear understanding of the project’s goals.

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Following this, general descriptions are presented to outline the system's role, user interactions, and constraints. The specific requirements section covers detailed functionalities, interfaces, and use cases essential for guiding the development and testing phases. Finally, analysis models are included to illustrate workflows and data handling processes. This organized structure ensures that the document serves as a reliable reference throughout the software development lifecycle.

2. General Description

The following section provides a high-level overview of the EDUassist system, outlining its purpose, primary functions, user characteristics, constraints, and dependencies. This description aims to contextualize EDUassist within the academic environment, emphasizing its role as a supportive tool for students and educators.

2.1 Product Perspective

EDUassist is an academic management tool created to assist students with efficient study planning, collaborative learning, and resource management. The platform seamlessly integrates with Google Calendar and Learning Management Systems (LMS), allowing users to synchronize and organize academic events and tasks without switching between multiple applications. EDUassist is designed as a supportive layer that enhances existing educational platforms rather than replacing them. The platform includes a set of AI-powered features, such as adaptive study recommendations and an interactive whiteboard, to provide users with personalized and dynamic learning aids that evolve based on individual academic progress.
The EDUassist platform is structured around a centralized interface accessible via a web application, with essential data stored in a secure database to ensure data privacy and reliability. This approach allows users to access and manage academic resources and collaborative tools in one unified space, improving their learning efficiency and academic engagement.

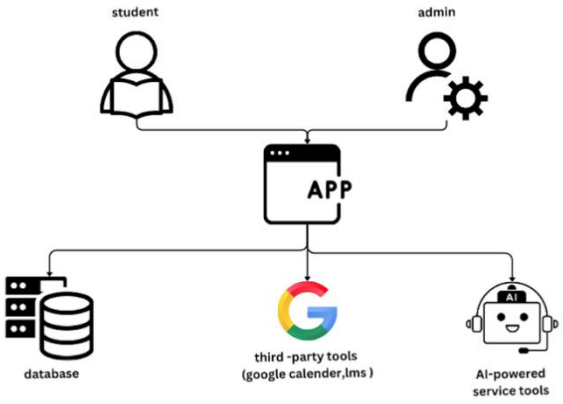


Figure 1 Product Perspective

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2.2 Product Functions

This subsection provides a summary of the functions that the EDUassist software will perform, categorized by the interactions of each actor: Admin, End User, and the System.

Admin (Developer):

- **User Management:**
  - Admins can create, modify, and delete user accounts for end users, ensuring proper access and permissions.
  - Admins can reset passwords and manage user roles to maintain security and functionality within the system.
- **System Monitoring:**
  - Admins can monitor system performance and usage statistics, allowing for better resource allocation and system improvements.
  - Admins can generate reports on user activity and engagement metrics to assess the effectiveness of the platform.
- **Configuration Settings:**
  - Admins can configure system settings, including integrations with external tools and services, optimizing the user experience.

End User:

- **Account Management:**
  - Users must sign in to access their personalized study plans and resources. Without signing in, they cannot utilize any features of the EDUassist software.
  - Each user has a personal account for easy sign-in and sign-out, with options for password recovery.
- **Personalized Study Planning:**
  - Users can generate customized study plans tailored to their individual academic goals and learning preferences.

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  - Users can track academic tasks and deadlines, helping them stay organized and focused.

- **AI-Assisted Features:**
  - Users can utilize an AI-assisted whiteboard to visualize complex concepts and collaborate in real time with peers.
  - Users can generate short explanatory videos from their whiteboard sessions, providing a dynamic way to revisit challenging topics.
- **Resource Access:**
  - Users receive personalized recommendations for study materials based on their progress and preferences.
- **Collaboration Tools:**
  - Users can share resources and collaborate on projects with peers through integrated communication tools, enhancing collaborative learning experiences.

System (EDUassist):

- **Authentication Services:**
  - The system provides secure authentication for all users, managing sign-in and sign-out processes effectively.
  - The system supports password recovery and two-factor authentication to enhance security.
- **Data Management:**
  - The system securely stores user profiles, study plans, and progress tracking data, ensuring data integrity and privacy.
  - The system maintains an up-to-date library of educational resources, facilitating easy access for users.
- **Integration Services:**
  - The system integrates with external calendar services, allowing users to manage study schedules and receive reminders.

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- The system facilitates communication between users for collaboration and resource sharing.

- **Performance Monitoring:**
  - The system tracks user interactions and generates usage statistics, providing valuable insights for admins to improve the platform.

- **AI Tools and Assistance:**
  - The system provides AI-powered features to enhance the user experience, such as real-time suggestions, personalized recommendations, and intelligent content generation.
  - The AI assistance is seamlessly integrated throughout the platform, supporting users in various tasks like whiteboard collaboration, study plan creation, and resource discovery.

2.3 User Characteristics

There are three types of users that interact with the EDUassist system: end users, admins, and EDUassist. Each of these user types has different interactions with the system, resulting in distinct requirements.

- **End Users:**
  End users include individuals aged 12 and older, such as middle school students, high school students, college students, and adult learners. Their primary interaction with the system is through the mobile application. End users can create personalized study plans based on syllabi they provide. They can input their course materials, including topics covered, deadlines, and specific learning objectives. The system will then generate tailored study plans that guide users through their academic responsibilities. To achieve this, end users need an intuitive interface that supports various learning styles and allows for easy input of syllabi. They also require features for tracking progress, receiving reminders for upcoming tasks, and accessing resources aligned with their study plans.

- **Admins (Developers):**
  Admins, who are the developers of EDUassist, interact with the system primarily through the software interface. Their main role is to configure tools and settings within the EDUassist platform to ensure optimal functionality. While they do not manage educational resources directly, they are responsible for maintaining the software environment, implementing updates, and troubleshooting issues that arise. Admins require tools for configuration, user

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management, and monitoring system performance to ensure that the platform runs smoothly and efficiently.

- **EDUassist:**
  EDUassist operates in the background, analyzing user interactions and providing intelligent features. It analyzes the syllabi submitted by end users and generates personalized study plans that align with the topics covered. EDUassist also assists in generating videos from whiteboard content and provides study recommendations based on the user's academic needs. To effectively support users, EDUassist must be capable of processing diverse data inputs, including syllabi, and delivering relevant suggestions, study plans, and video content. Additionally, it should adapt to user feedback to improve its recommendations over time.

2.4 General Constraints

This section outlines the general limitations and constraints that must be considered during the theoretical design of the EDUassist system:

- **Documentation Scope:**
  The project focuses solely on documentation and theoretical design. No physical implementation or prototype development will take place, limiting practical validation of the proposed design.

- **Budget Constraints:**
  The theoretical design assumes a limited budget for implementation, which influences assumptions about hardware, software, and resources.

- **Technological Assumptions:**
  The design is constrained to assume compatibility with widely available tools like Google Calendar, Zoom, and Microsoft Teams, dictating the integration requirements and task management workflows.

- **Resource Availability:**
  Since the project involves a small team with fixed expertise, the design cannot incorporate features that would require advanced development skills or additional technical training.

- **Scalability Limitations:**
  Scalability and performance enhancements are theoretical and untested. The design will assume scalability requirements based on estimated future user needs.

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- **Security Protocols:**
  The design must include strong security assumptions, such as data encryption and multi-factor authentication, to theoretically meet user data protection requirements.

- **Network Dependency:**
  Features requiring cloud functionality assume stable and reliable internet connections. The design does not account for offline capabilities.

- **Time Constraints:**
  The project documentation must be completed within the predefined timeline, which restricts the depth of theoretical exploration.

- **Language Support:**
  The system is designed to theoretically support multiple languages, including Arabic, but the scope does not extend to detailed linguistic or cultural customizations.

2.5 Assumptions and Dependencies

The assumptions and dependencies that could affect the EDUassist SRS requirements are listed in this subsection. Any modifications to these elements may call for modifying the specified requirements.

- **Assumptions**

  - Availability of Required Hardware and Software: It is anticipated that students will have access to gadgets that are compatible with current operating systems (Windows 10+, macOS, iOS, Android) and web browsers (Chrome, Safari, Edge), such as laptops and smartphones.

  - Integration with Third-Party Tools: In order to achieve integration features, the system design makes the assumption that third-party tools like Microsoft Teams, Zoom, and Google Calendar will remain available and functional.

  - Reliable Internet Connectivity:
    The platform assumes that users will have stable and reliable internet access to fully utilize cloud-based features such as collaborative whiteboards and real-time data syncing.
  - Static User Base Growth:
    It is assumed that the growth in the user base during the initial years will remain manageable within the designed scalability limits.

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- Resource Availability for Future Implementation:
  It is assumed that if the project moves from being purely documentation-based to actual implementation, the required resources—whether financial, technical, or human—will be accessible to support the process.

- Regulatory Environment:
  It is assumed that applicable data privacy and protection laws, such as GDPR, will remain consistent without introducing additional requirements that might affect system design.

- **Dependencies**

  - Operating Systems:
    The system depends on the availability of up-to-date versions of supported operating systems. Any changes or discontinuations in these OSs could necessitate updates to system requirements.

  - Technological Ecosystem:
    The project depends on the compatibility of its design with current cloud technologies, API standards, and integration protocols. Any significant changes to these technologies may affect the feasibility of integration.

  - User Feedback Mechanism:
    The design assumes that user feedback will be systematically gathered during testing phases. Dependence on feedback mechanisms means potential limitations in refining the theoretical design.

  - Platform Accessibility Standards:
    We're assuming that the current accessibility standards, like WCAG 2.1, will stay the same. If these standards change, we might need to adjust the system's requirements to match.

  - Educational Policies.
    The project assumes that the current curriculum structure and educational methodologies will remain relatively stable. Major shifts in these could affect system requirements, particularly for the study plan feature.

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3. Specific Requirements

This section outlines the software requirements that will guide the project. Each requirement must adhere to principles of correctness, traceability, unambiguity, and verifiability. The requirements presented here will serve as the foundation for design decisions, without prematurely constraining the implementation. Careful structuring of this section is crucial, as it will be the primary reference point moving forward.

3.1 External Interface Requirements

3.1.1 User Interfaces

3.1.1. Common Interfaces:

3.1.1.1. Sign In Interface

Sign In Interface	
Field	User ID, Password
Format	Encrypted Text
Level	Required
Input/Output	Input
Comment	Unique identifier for each user, Must be at least 8 characters, containing numbers, uppercase and lowercase letters, and special characters.

Table 6 Sign In Interface

3.1.1.2. Forget My Password Interface

Forget My Password Interface	
Field	Email, User ID
Format	Text
Level	Required
Input/Output	Input
Comment	Email must match the registered email in the database, User ID is the unique identifier for password recovery.

Table 7 Forget My Password Interface

3.1.1.3. Settings Interface

Settings Interface	
Field	Personal Information, Notification Preferences, Privacy Settings, Application Preferences, Account Management
Format	Text, Toggle, Toggle, Dropdown, Button
Level	Input/Output
Input/Output	Input/Output
Comment	Enables users to customize their profile, manage notifications, control privacy, and adjust application settings.

Table 8 Settings Interface

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3.1.1.4. Login Interface

Login Interface	
Field	User ID, Password
Format	Encrypted Text
Level	Required
Input/Output	Input
Comment	Unique identifier for each user; Password must be at least 8 characters and include numbers, uppercase and lowercase letters, and special characters.

Table 9 Login Interface

3.1.2. Admin Interfaces

3.1.2.1. Admin Homepage Interface

Admin Homepage Interface	
Field	Manage Users, View Reports, Notifications
Format	Button, Button, List
Level	Required
Input/Output	Output
Comment	Central dashboard for accessing user management, reports, and notifications.

Table 10 admin Homepage Interface

3.1.2.2. User Management Interface

User Management Interface	
Field	User List, Edit Button, Delete Button
Format	Table, Button, Button
Level	Required
Input/Output	Input/Output
Comment	Displays registered users with options to edit or delete user details

Table 11 User Management Interface

3.1.2.3. Reports Generation Interface

Reports Generation Interface	
Field	Report Type, Generate Button
Format	Dropdown, Button
Level	Required
Input/Output	Output
Comment	Allows administrators to select report types and generate performance and activity reports.

Table 12 Reports Generation Interface

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3.1.2.4. Notifications Interface

Notifications Interface	
Field	Notification List, Alert Settings
Format	List, Toggle
Level	Required
Input/Output	Output/Input
Comment	Displays system alerts and allows admins to manage notification preferences.

Table 13 Notifications Interface

3.1.3. End user

3.1.3.1. Dashboard Interface

Dashboard Interface	
Field	Notifications, Quick Access to Features
Format	List, Buttons
Level	Required
Input/Output	Output
Comment	Central hub for notifications and quick access to main features such as the whiteboard and AI tools.

Table 14 Dashboard Interface

3.1.3.2. Task and Assignment Tracker Interface

Task and Assignment Tracker Interface	
Field	Task List, Assignment List, Progress Indicators, Deadlines, Status
Format	Table, Table, Progress Bar, Date, Dropdown
Level	Required
Input/Output	Output
Comment	Monitors tasks and assignments, showing their progress, deadlines, and current status in one unified interface.

Table 15 Task and Assignment Tracker Interface

3.1.3.3. Calendar Integration interface

Calendar Integration	
Field	Google Calendar
Format	Calendar
Level	Required
Input/Output	Input/Output
Comment	Integrates Google Calendar for managing tasks and deadlines associated with projects.

Table 16 Calendar Integration Interface

3.1.3.4. Profile Interface

Profile Interface

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Field	Full Name, Email, Preferences
Format	Text, Text, Dropdown
Level	Required
Input/Output	Input
Comment	Enables users to view and edit their profile information .

Table 17 profile Interface

3.1.3.5. Assignment Tracker Interface

Assignment Tracker Interface	
Field	Full Name, Email, Preferences
Format	Text, Text, Dropdown
Level	Required
Input/Output	Input
Comment	Enables users to view and edit their profile information .

Table 18 Assignment Tracker Interface

3.1.3.6. Whiteboard Interface

Whiteboard Interface	
Field	Drawing Tools, Chat for Video Generation, Resource Suggestions, Collaboration Tools
Format	Canvas, Text Box, List, Buttons
Level	Required
Input/Output	Input/Output
Comment	Central feature for collaborative brainstorming, includes a chat for generating videos or recommendations, with resources stored in the library. Allows users to share content directly within the Whiteboard interface for collaborative projects.

Table 19 Whiteboard Interface

3.1.3.7. Video Generation Interface (Sub-interface of Whiteboard)

Video Generation Interface (Sub-interface of Whiteboard)	
Field	Text Input, Generate Video Button
Format	Text Box, Button
Level	Required
Input/Output	Input/Output
Comment	Utilizes AI to convert text into videos for enhancing understanding of concepts, accessible from the Whiteboard.

Table 20 Video Generation Interface (Sub-interface of Whiteboard)

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3.1.3.8. Resource Library Interface and Personalized Study Plans interface

Resource Library Interface and Personalized Study Plans interface	
Field	AI Resource Recommendations, Study Plans
Format	List, Plan
Level	Required
Input/Output	Output
Comment	Provides access to personalized AI-generated resources and study plans tailored to student needs, with recommendations displayed in the Whiteboard.

Table 21 Resource Library Interface and Personalized Study Plans interface

3.1.2 Hardware Interface

The EDUassist platform is purely a software-based solution, so it doesn't require specific hardware interfaces. Instead, it depends on the device's operating system to handle tasks like connecting to the database server and other services. All user interactions happen through the software's interface, making hardware dependencies minimal and indirect.

3.1.3 Software Interfaces

The EDUassist application is designed to operate on a range of mobile devices, supporting iOS 12.4.1 and later, as well as Android 8.1 (Oreo) and newer. While these versions ensure compatibility with many devices, EDUassist functions optimally on more recent versions, such as iOS 17 and Android 13 or 14, leveraging the latest security and performance features. The application connects to a MySQL database server for securely managing user data, including profiles, study plans, and resources. For secure sign-in, an OTP (One-Time Password) Generation Server provides temporary passwords, enhancing login security. Additionally, EDUassist integrates with a mail server for automated password recovery and notifications, facilitating important user communication.

3.1.4 Communications Interfaces

The EDUassist platform uses several communication methods to make sure users can interact with the system easily, securely, and efficiently.
These methods also help the system work with other tools like Google Calendar and Learning Management Systems (LMS) to give users a smooth experience.

1. **User-to-System Communication**

- Purpose: This allows students and admins to interact with the EDUassist system, such as logging in, uploading files, or accessing their study plans.
- Protocols: The system uses HTTPS for secure communication and WebSocket for real-time updates.
- How It Works: When users perform actions like logging in or uploading a file, the system processes the request and responds with the necessary information, like updated tasks or a study plan.

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- Example: like when a student uploads their course syllabus, and EDUassist generates a personalized study plan based on the file.

2. System-to-External Tools Integration

- Purpose: This makes it possible for EDUassist to sync data with tools like Google Calendar and LMS platforms.
- Protocols: The system uses REST APIs to exchange data and OAuth 2.0 to securely manage user authentication.
- How It Works: EDUassist connects to external tools using secure API connections to send or receive data, like updating events or deadlines.
- Example: like when a student's study schedule is synced with their Google Calendar, so they get reminders for tasks and deadlines.

3. System-to-User Notifications

- Purpose: The system sends reminders and updates to users about tasks, deadlines, or any important changes.
- Protocols: Email notifications are sent using SMTP, while mobile notifications use push notification services like Firebase.
- How It Works: The system detects important events (like a deadline) and sends alerts to the user's email or mobile app.
- Example: When EDUassist sends a notification to remind a student about an upcoming assignment deadline.

4. System-to-System Communication

- Purpose: This is how different parts of EDUassist (like the database, AI tools, and user interface) work together to handle tasks and provide features.
- Protocols: The platform uses internal APIs and message queues (like RabbitMQ) to share data efficiently.
- How It Works: When a user performs an action, the system's components communicate in the background to process the request, like fetching data or generating a study plan.
- Example: like when the AI engine analyzes a user's input and provides study recommendations instantly.

By designing these interfaces, EDUassist ensures secure, connected, real-time, and reliable communication, enabling seamless interactions between users and the system, as well as synchronization with external tools.

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3.2 Functional Requirements

The following section describes the functional requirements of the EDUassist project, outlining the key features of the system. Each feature is detailed with inputs, processing, outputs, and error handling.

3.2.1 User Registration and Authentication

3.2.1.1 Introduction	Enables students and admin to register and securely log in, while the system manage the authentication process.
3.2.1.2 Input	• Admin username, email address, password • Student username, email address, password
3.2.1.3 Processing	• Validate input fields for completeness and correctness. • Encrypt passwords before storage. • Authenticate users during login using stored credentials.
3.2.1.4 Output	• Admin Will be able to access the system interface where they can manage platform and users accounts • Student Will be able to access the dashboard interface where they can user the other many functionalities
3.2.1.5 Error handling	• Provide descriptive error messages for incorrect login attempts. • Lock accounts after multiple failed attempts. • Notify admins of suspicious login attempts.

Table 22 User Registration and Authentication

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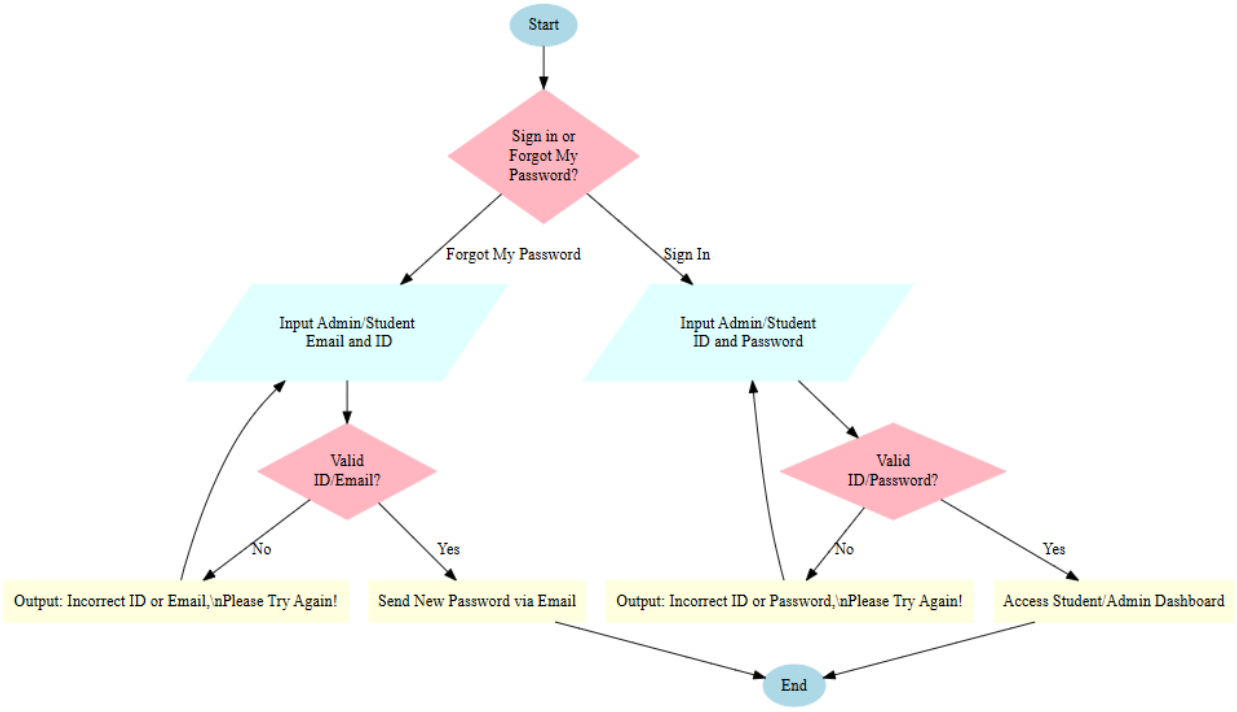


Figure 2 User Registration and Authentication

3.2.2 AI- assist Whiteboard

3.2.2.1 Introduction	A virtual shared whiteboard designed for real-time student collaboration. The system analyzes user interactions and preferences on the whiteboard to dynamically suggest relevant learning resources.
3.2.2.2 Input	Student actions: Add text, images, or drawings. Share and collaborate with peers. Add external Resources : slides , reference book…etc.
3.2.2.3 Processing	Synchronize whiteboard actions across all participants in real-time. Temporarily store session data for retrieval and export. Match ongoing activities with user profile data to identify learning gaps. Retrieve relevant study materials from internal and external sources.
3.2.2.4 Output	Updated whiteboard view for all participants. Exported content (PDF or image format). Identify learning needs, correct inaccuracies, and suggest missing details.
3.2.2.5 Error handling	Retry synchronization in case of network delays. Notify users of session errors. Provide default suggestions if specific recommendations are unavailable.

Table 23 AI- assist Whiteboard

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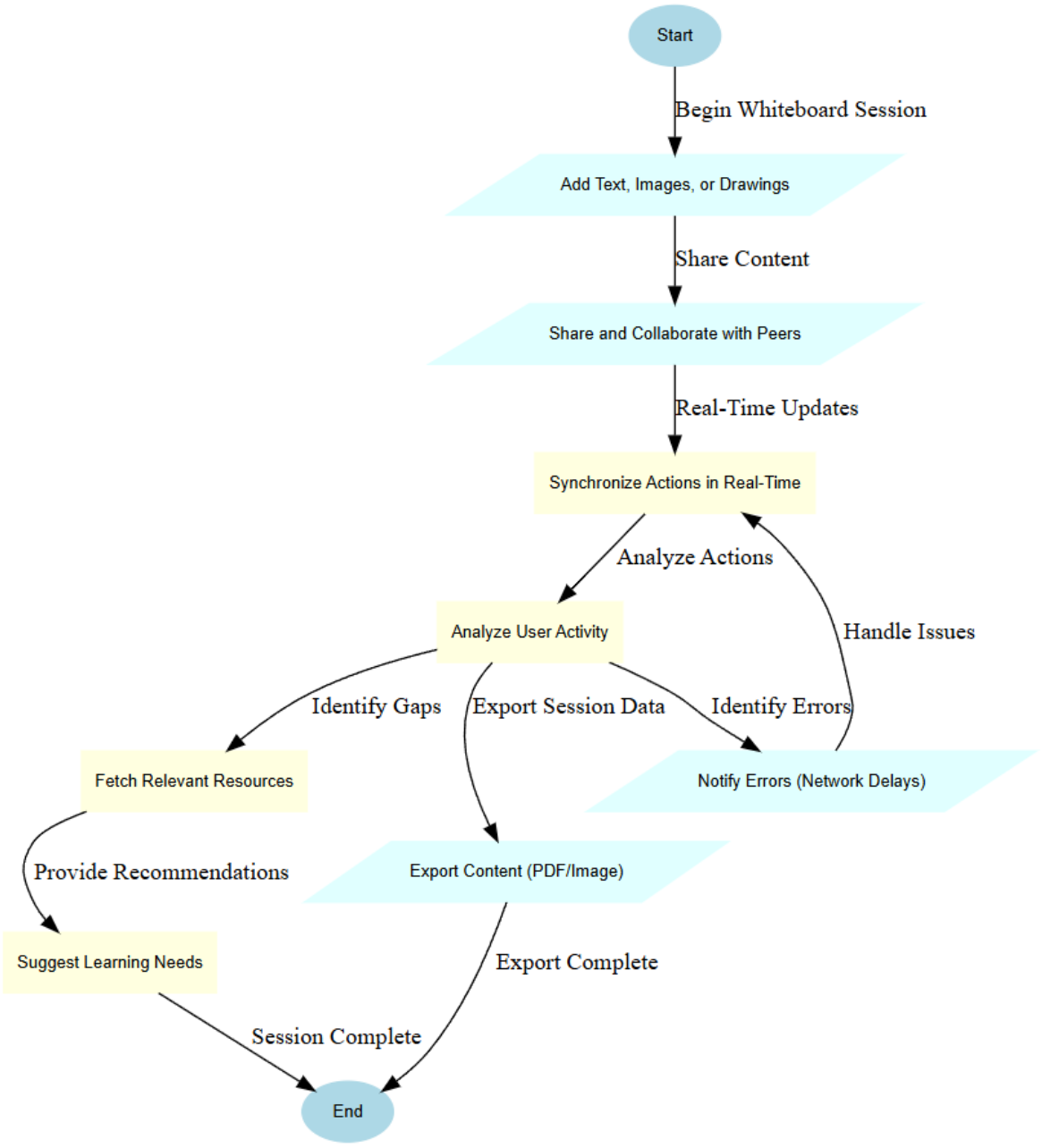


Figure 3 AI- assist Whiteboard

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3.2.3 Study Plan Management

3.2.3.1 Introduction	Allows students to create, view, and modify personalized study plans tailored to their academic goals.
3.2.3.2 Input	Syllabus file (uploaded by the student). Preferences (e.g., focus areas, deadlines, daily study hours).
3.2.3.3 Processing	Parse syllabus files to extract topics. Match user preferences with syllabus data. Generate a structured study plan.
3.2.3.4 Output	Personalized study plans displayed to students . Alerts for missed or upcoming tasks. Sync with external tools like Google Calendar.
3.2.3.5 Error handling	Notify users if the syllabus file format is unsupported. Fallback to a default study plan if parsing fails.

Table 24 Study Plan Management

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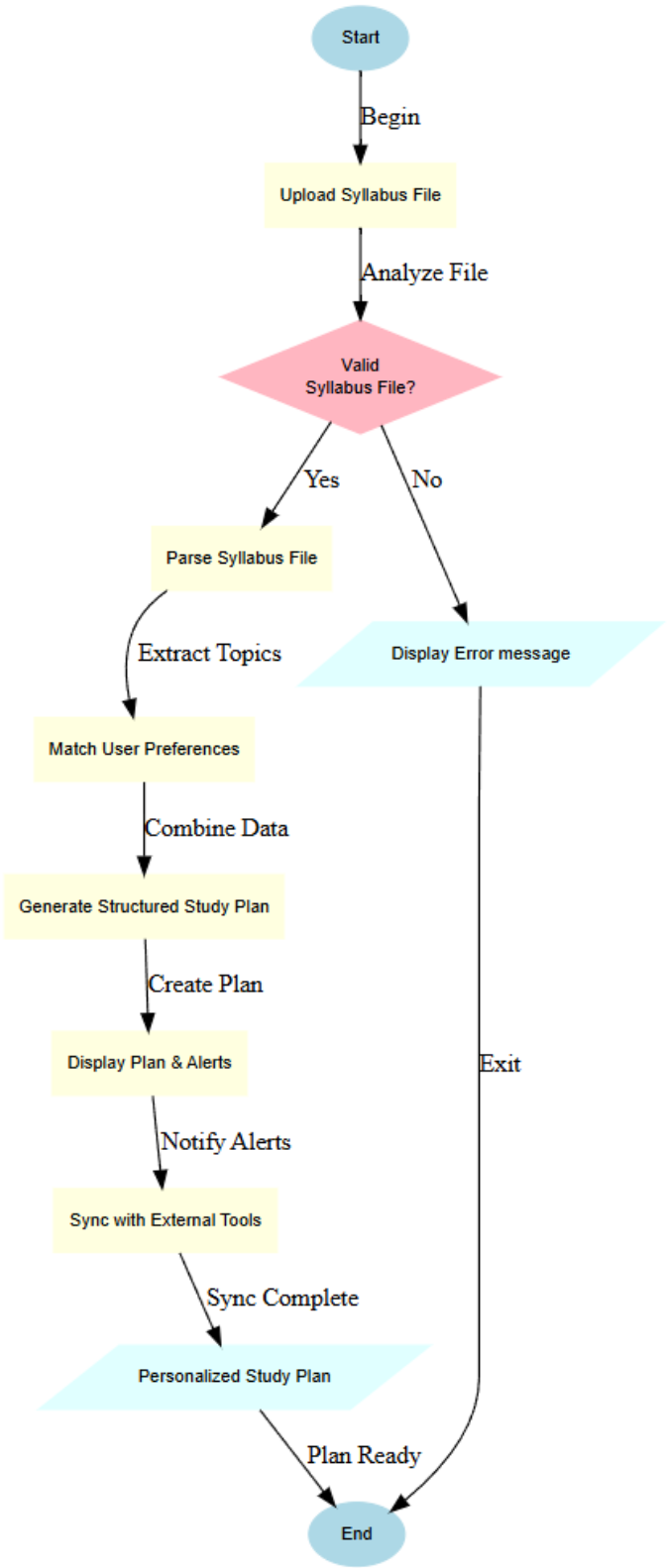


Figure 4Study Plan Management

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3.2.4 Resource Recommendations

3.2.4.1 Introduction	AI-driven feature to recommend study materials (videos, articles, books) based on user academic needs.
3.2.4.2 Input	User profile (academic level, courses, preferences). Progress tracking data. Course-specific requirements
3.2.4.3 Processing	Match student needs with available resources. Prioritize resources based on relevance.
3.2.4.4 Output	Recommendations displayed on the dashboard.
3.2.4.5 Error handling	Notify users if recommendations are unavailable. Display fallback suggestions.

Table 25 Resource Recommendations

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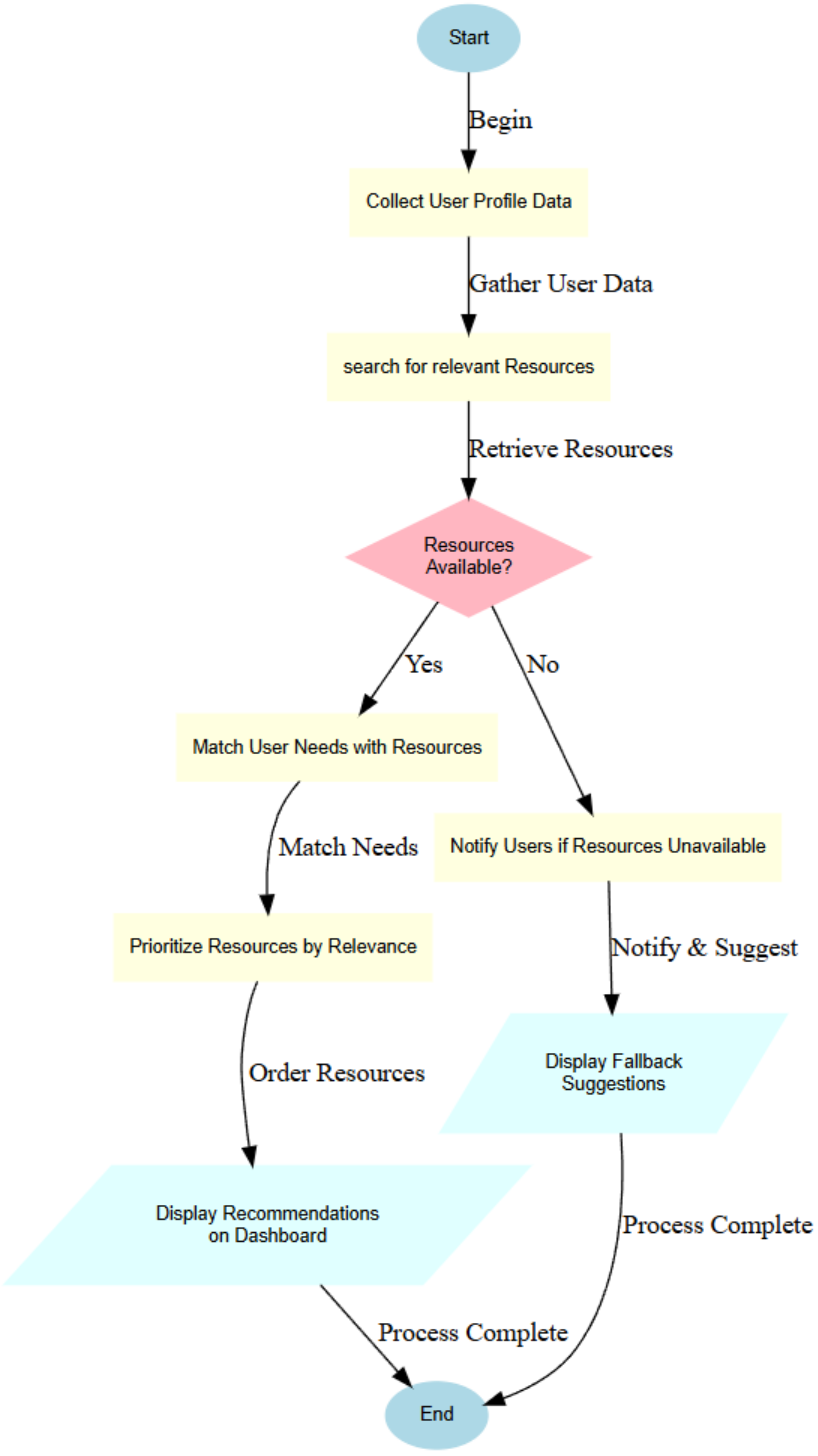


Figure 5Resource Recommendations

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3.2.5 Task Tracking

3.2.5.1 Introduction	Monitors task progress for students and provides visual updates.
3.2.5.2 Input	Task details from users (e.g., deadlines, progress updates). Calendar integration data.
3.2.5.3 Processing	Update task statuses based on user input. Sync with external tools like Google Calendar.
3.2.5.4 Output	Visualized task progress. Notifications for overdue tasks.
3.2.5.5 Error handling	Retry synchronization with external tools on failure. Notify users of incomplete or missing task updates.

Table 26 Task Tracking

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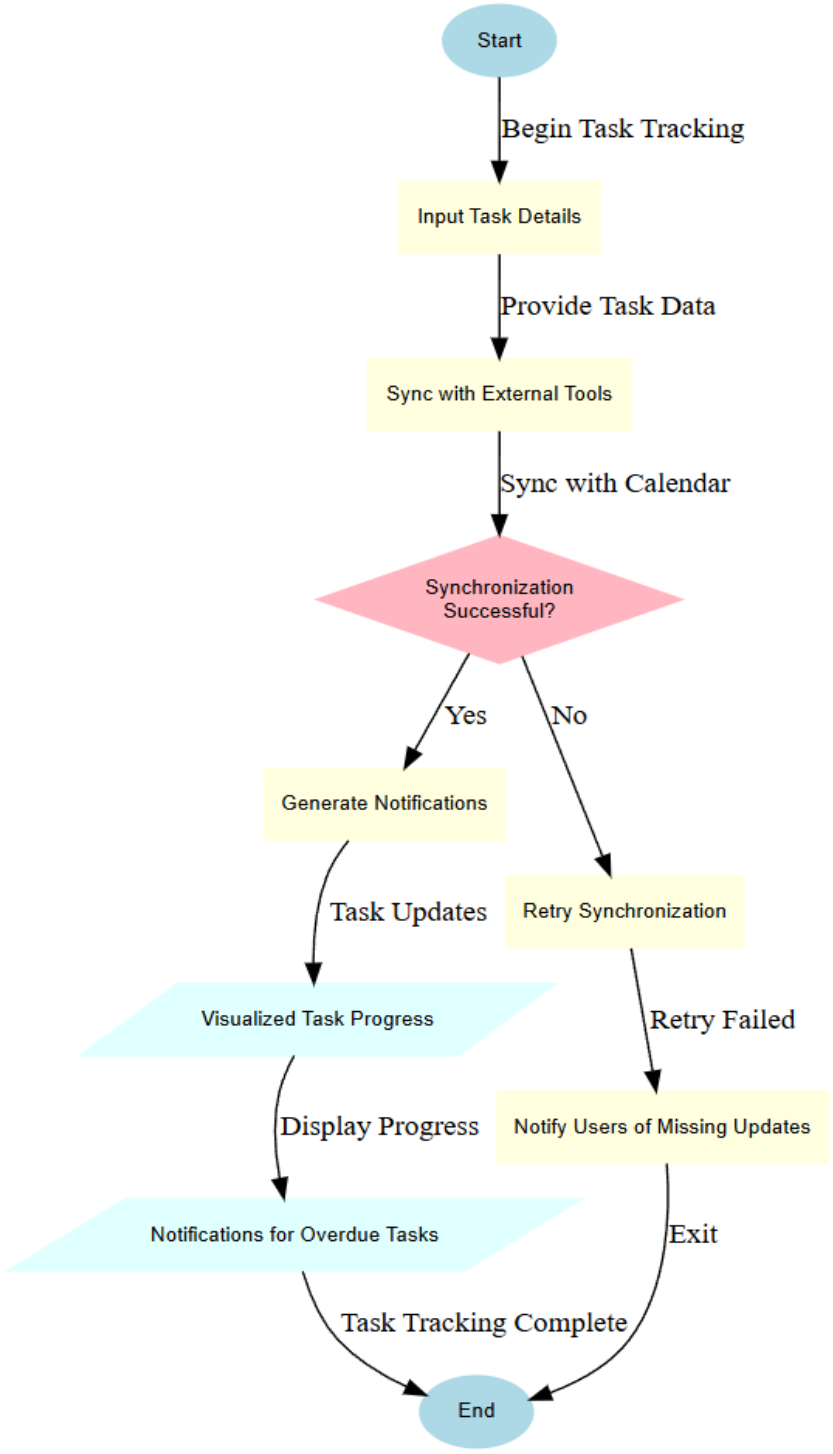


Figure 6Task Tracking

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3.2.6 Visual Representation of Text

3.2.6.1 Introduction	Converts textual content into visual formats such as charts, graphs, or mind maps for enhanced learning.
3.2.6.2 Input	Text content uploaded by students. External sources of text (e.g., study guides).
3.2.6.3 Processing	Parse and analyze the provided text. Generate visual representations based on user preferences or text type.
3.2.6.4 Output	Visualized content displayed on the dashboard or exported as videos or gifs.
3.2.6.5 Error handling	Notify users if the text format is unsupported. Suggest modifications for better visualization results.

Table 27 Visual Representation of Text

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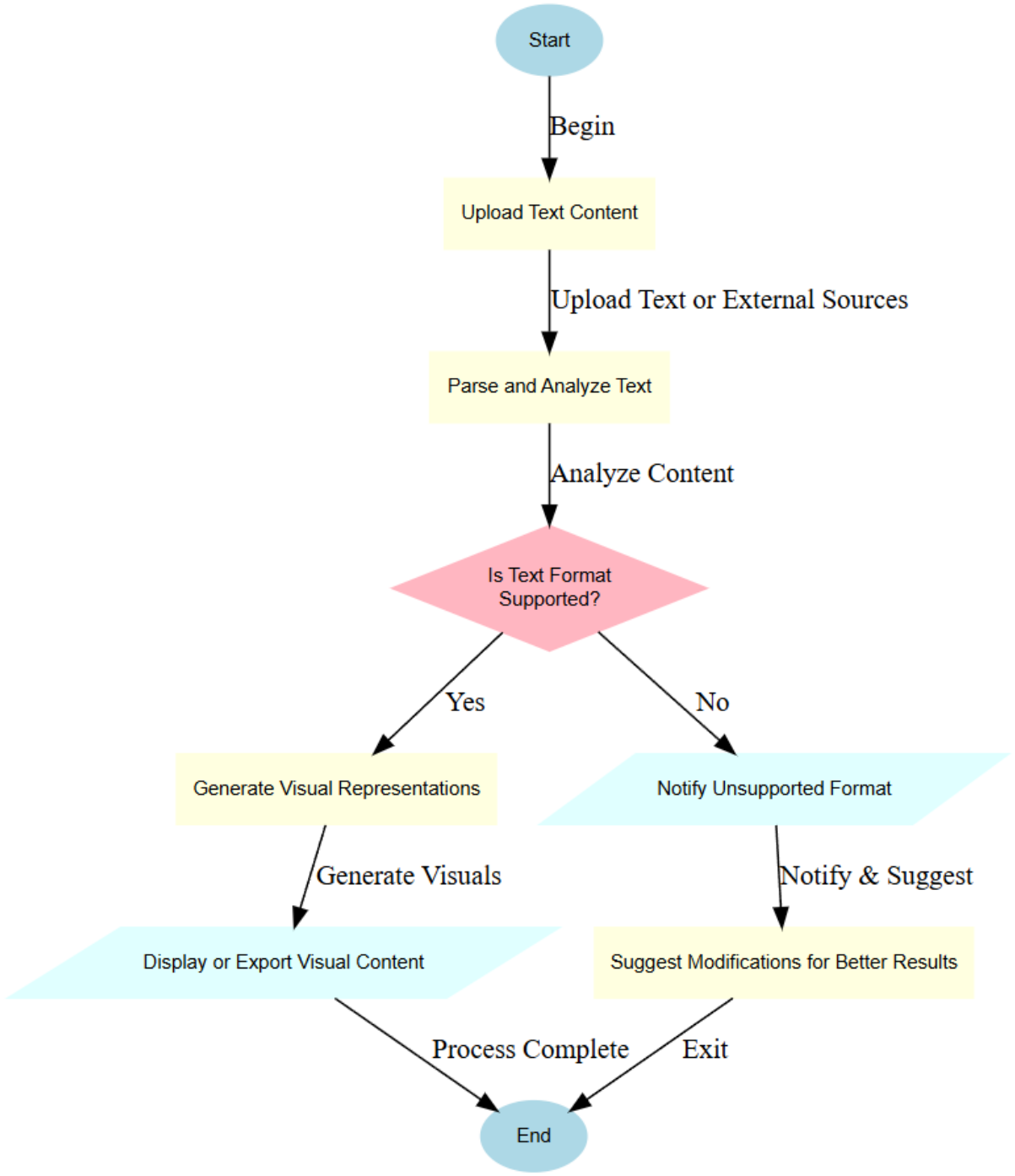


Figure 7Visual Representation of Text

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3.2.7 User Management

3.2.7.1 Introduction	Enables administrators to create, update, and delete user accounts, reset passwords, and manage roles to ensure appropriate access
3.2.7.2 Input	User details: Username, email address, password, and role. Search criteria for updating, creating or deleting a user.
3.2.7.3 Processing	Validate input fields for completeness and correctness. Encrypt passwords before saving. Verify unique username/email for new accounts. Update or delete user accounts based on search criteria.
3.2.7.4 Output	Confirmation message for successful operations (e.g., account created, updated, or deleted).
3.2.7.5 Error handling	Display error if mandatory fields are missing. Notify admin if the username/email is already taken. Return an error message for invalid search criteria.

Table 28 User Management

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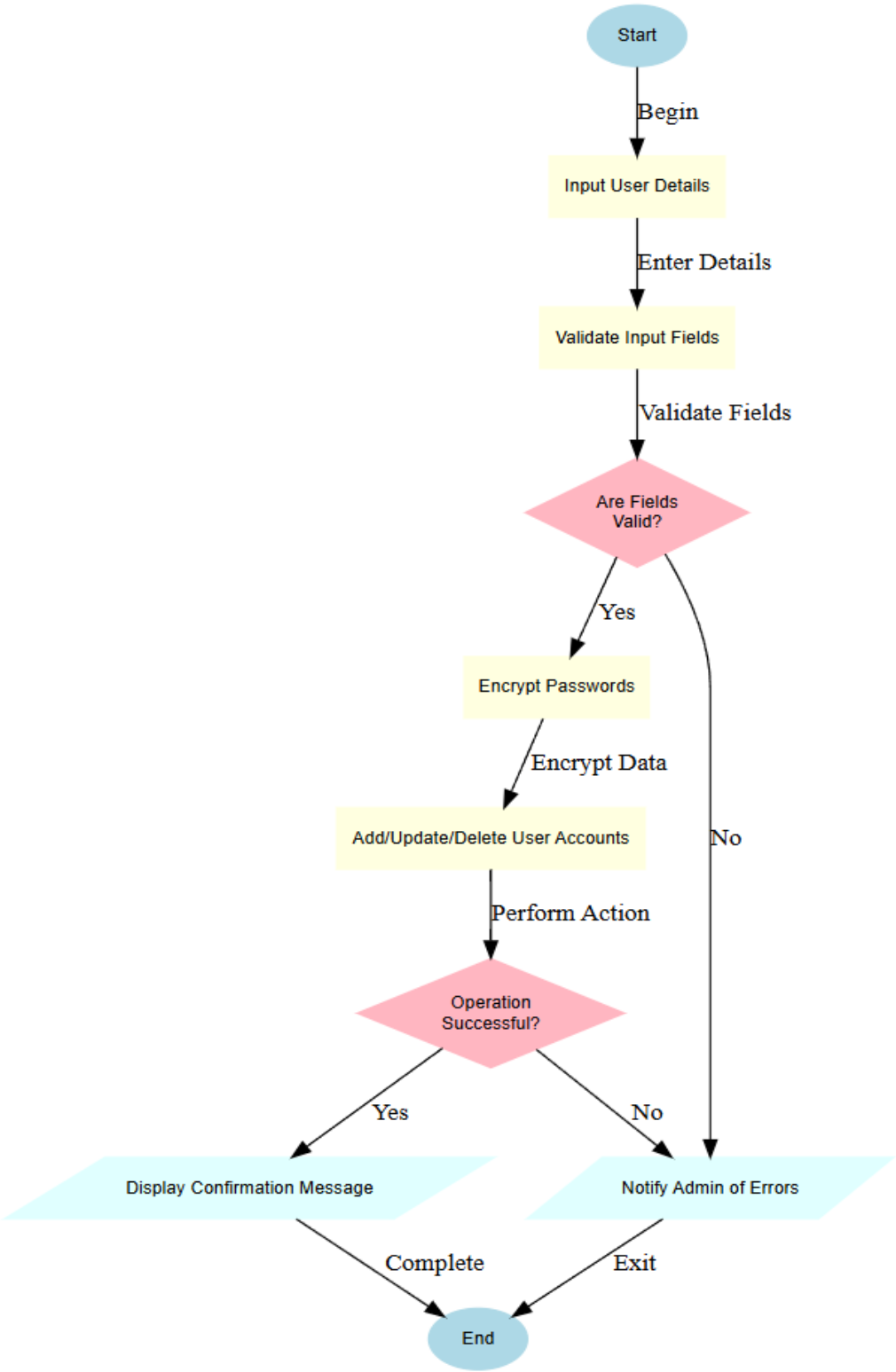


Figure 8User Management

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3.2.8 System Monitoring

3.2.8.1 Introduction	Enables administrators to monitor platform performance, usage statistics, and generate activity reports.
3.2.8.2 Input	Report type and time frame
3.2.8.3 Processing	Collect system usage data based on the specified report type and time frame. collect and format the data into a readable report.
3.2.8.4 Output	Generated report displaying user activity, resource utilization, or performance metrics.
3.2.8.5 Error handling	Notify admin if there's insufficient data for the selected time frame. Return an error if the report generation fails.

Table 29 System Monitoring

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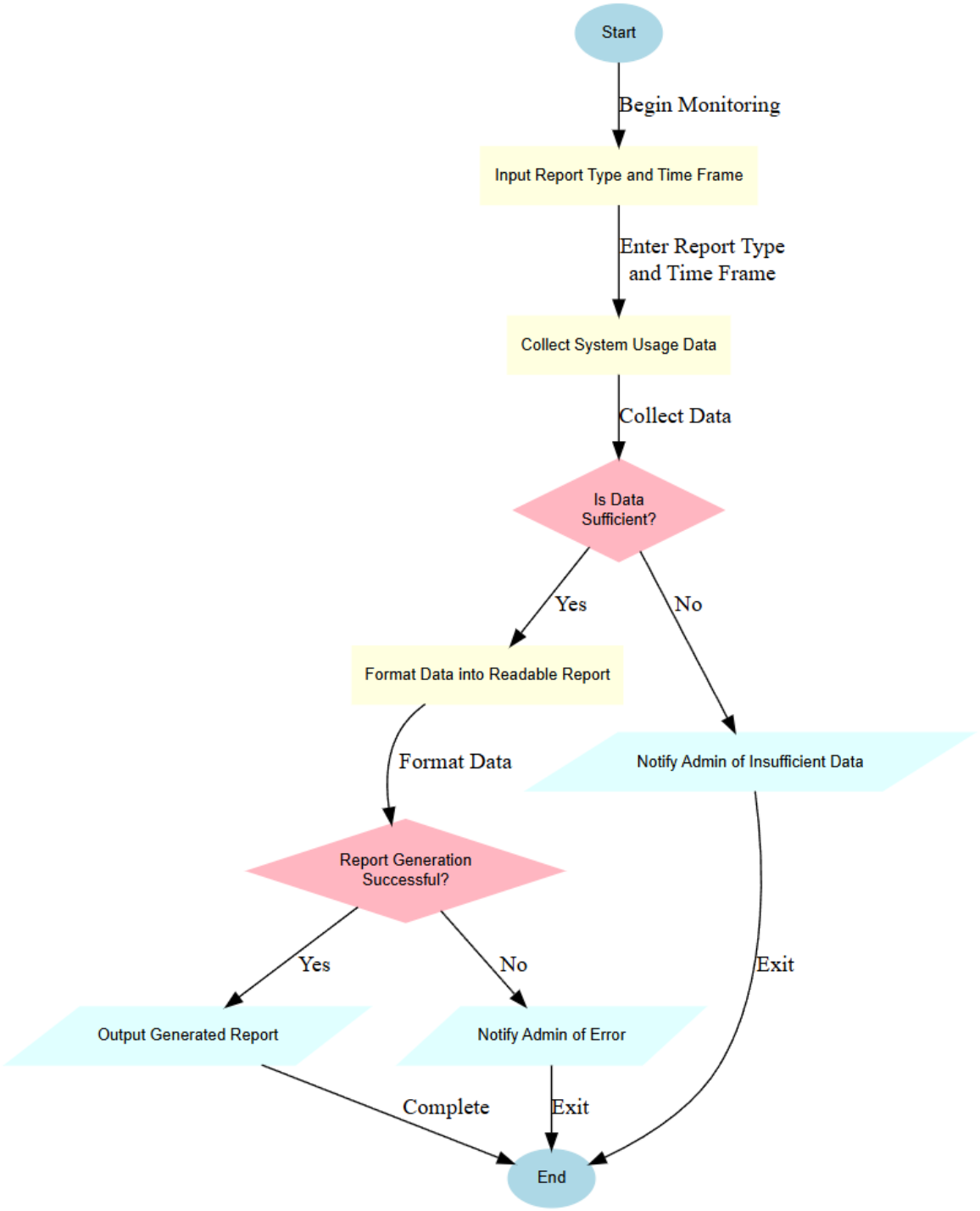


Figure 9 System Monitoring

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3.2.9 Configuration Settings

3.2.9.1 Introduction	Allows administrators to configure system settings and integrate external tools (e.g., Google Calendar).
3.2.9.2 Input	Settings to update (e.g., notification preferences, tool integrations). API keys or credentials for external tool integration.
3.2.9.3 Processing	Validate the input settings. Test and establish integration with external tools. Save the updated settings in the system configuration
3.2.9.4 Output	Confirmation message for successful configuration.
3.2.9.5 Error handling	Notify admin of invalid API keys or failed integration tests. Return an error if mandatory configuration fields are incomplete.

Table 30 Configuration Settings

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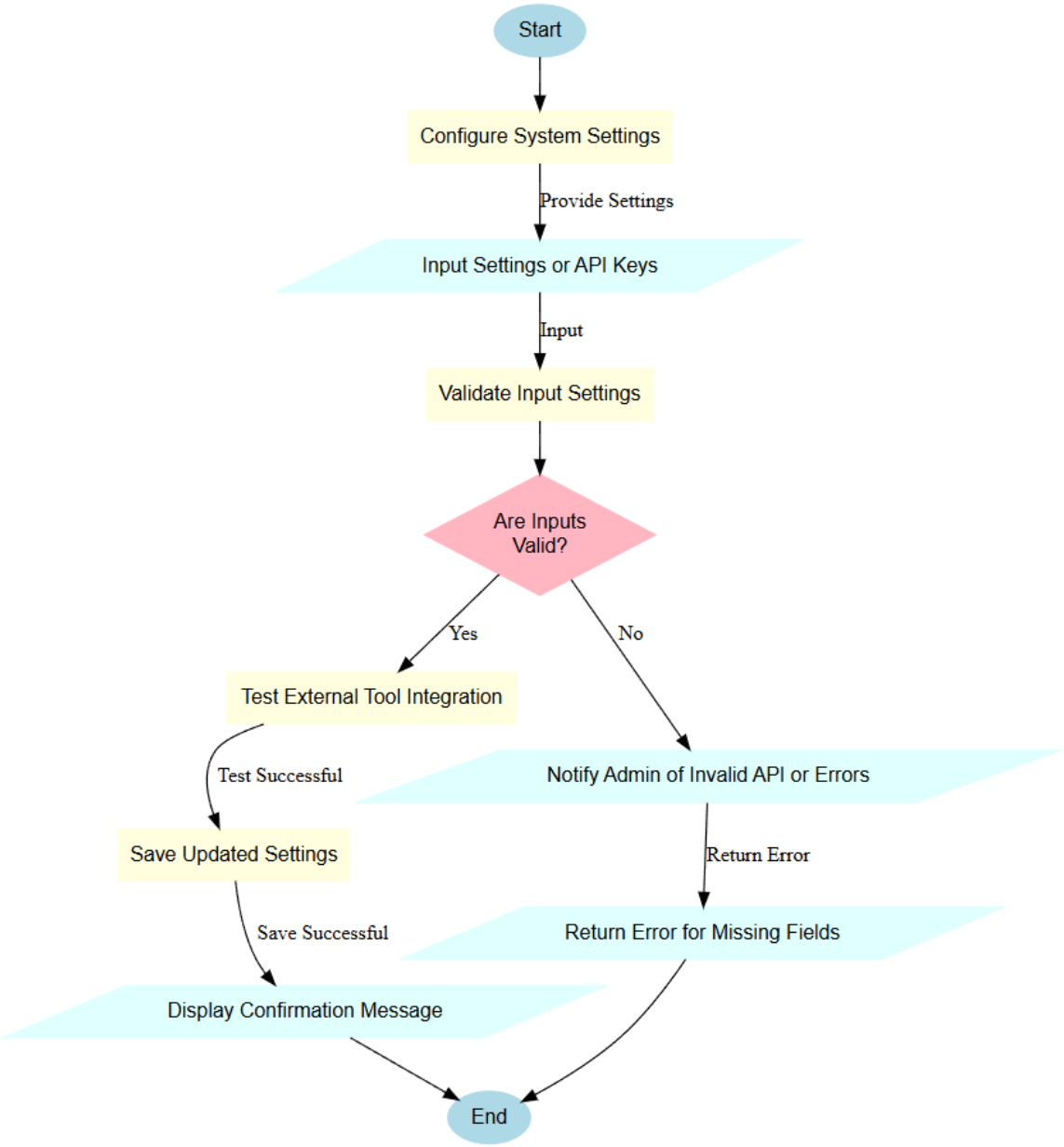


Figure 10Configuration Settings

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3.3 Use Cases

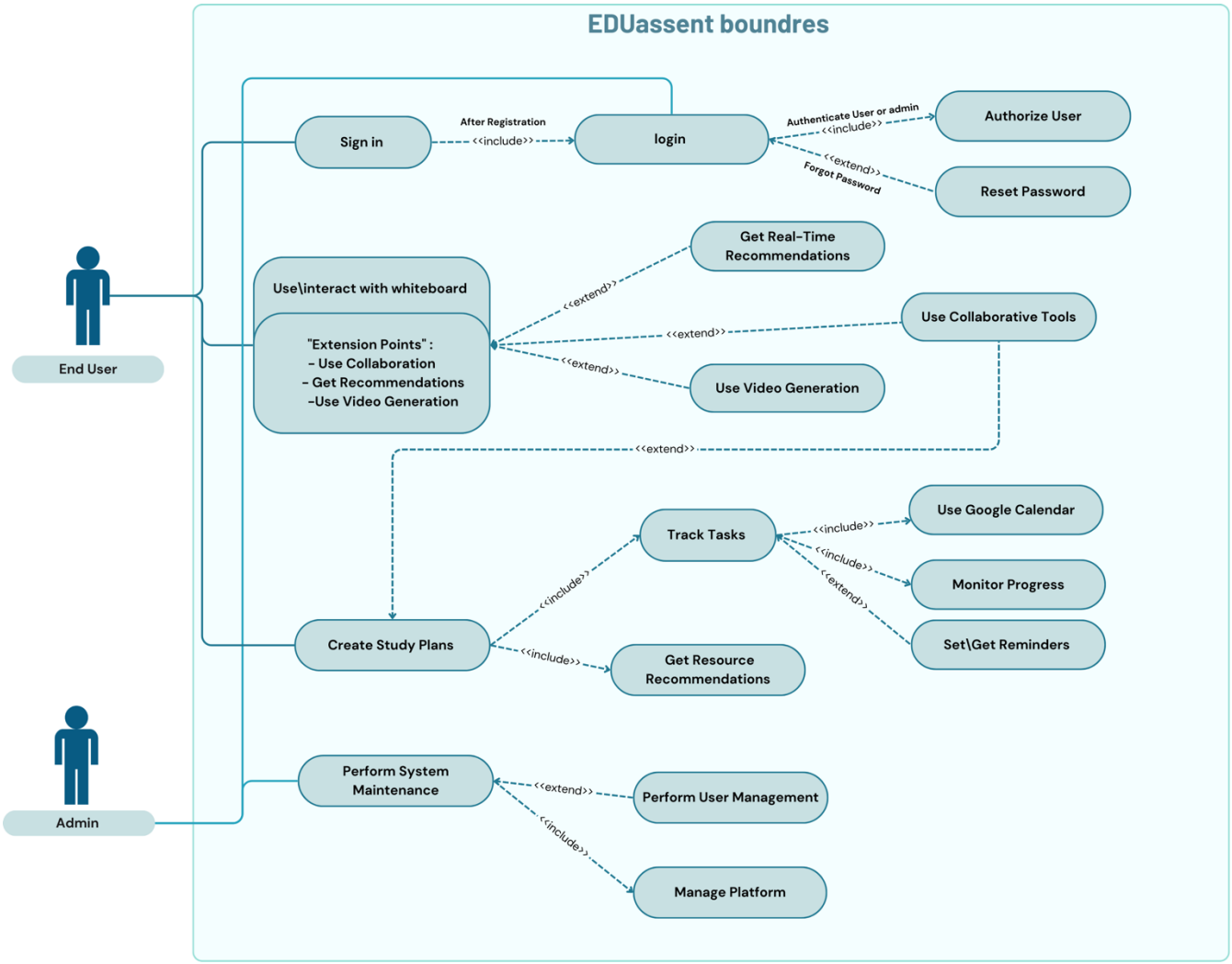


Figure 11: EDUassent Use case

This section outlines the various use cases of the application. Each use case provides insights into how users interact with the system and the specific features.

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3.3.1 Use Case: Login

Attribute	Details
Actors	End User, Admin
Description	Allows users to log in to the platform. The user can access the interactive whiteboard to annotate and visualize concepts.
Priority	High High priority
Data	Username, Password Whiteboard interface, drawing tools, text tools, and session data.
Stimulus	User clicks "Login" button. The user clicks on the "Interactive Whiteboard" button.
Response	User is authenticated and granted access. The whiteboard interface opens. - user can annotate or draw on the board. - Any drawings or text entered are saved in the session data.
Abnormal Conditions	If invalid credentials are entered, an error message is displayed. If the whiteboard fails to load, an error message is displayed, and the user is prompted to try again or contact support.

Table 31 Use Case: Login

3.3.2 Use Case: Sign In

Attribute	Details
Actors	End User
Description	Allows users to sign in to the platform.
Priority	High
Data	End user username, Email
Stimulus	User clicks "Sign In" button.
Response	User is authenticated and granted access.
Abnormal Conditions	If invalid ID or email is entered, an error message is displayed.

Table 32 Use Case: Sign In

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3.3.3 Use Case: Perform User Management

Attribute	Details
Actors	Admin
Description	Allows the admin to manage user accounts, including creating, updating, and deleting users. Allows users to sign in to the platform.
Priority	High
Data	User details (e.g., name, email, role) Admin/user ID, Email
Stimulus	Admin accesses the User Management module. User clicks "Sign In" button.
Response	Admin can perform CRUD (Create, Read, Update, Delete) operations on user accounts, including adding and removing users. User is authenticated and granted access.
Abnormal Conditions	- If the email address is already in use during user creation - If the user ID does not exist during deletion. - If invalid data is entered (e.g., incorrect format) If invalid ID or email is entered, an error message is displayed.

Table 33 Use Case: Perform User Management

3.3.4 Use Case: Manage Platform

Attribute	Details
Actors	Admin
Description	Allows the admin to manage the overall platform, including system configurations, settings, configuration of tools, and fixing issues.
Priority	High
Data	Platform-level data, configurations, tool settings, issue reports
Stimulus	Admin accesses the Manage Platform module.
Response	Admin can perform various administrative tasks, including configuring tools, fixing issues, and optimizing the platform.
Abnormal Conditions	- If invalid settings are provided during tool configuration. - If the admin lacks sufficient permissions to perform certain actions. - If there is a failure in saving changes due to server issues. - If an issue cannot be resolved due to lack of information or resources.

Table 34 Use Case: Manage Platform

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3.3.5 Use Case: Perform System Maintenance

Attribute	Details
Actors	Admin
Description	Allows the admin to perform system maintenance tasks, such as software updates, backups, troubleshooting, and fixing issues.
Priority	High
Data	System-level data, logs, maintenance reports
Stimulus	Admin accesses the System Maintenance module.
Response	Admin can execute maintenance procedures, perform updates, and fix identified issues to ensure smooth operation of the platform.
Abnormal Conditions	- If a software update fails due to compatibility issues. - If backup processes are interrupted or fail. - If logs indicate critical errors that cannot be resolved. - If the admin lacks permissions to perform certain maintenance tasks

Table 35 Use Case: Perform System Maintenance

3.3.6 Use Case: Reset Password

Attribute	Details
Actors	End User, Admin
Description	Allows users to manage their passwords, including resetting and updating them.
Priority	High
Data	Username, Email, New Password
Stimulus	User clicks "Forget My Password" button.
Response	An email is sent containing the new password.
Abnormal Conditions	If an invalid ID or email is entered, an error message is displayed.

Table 36 Use Case: Reset Password

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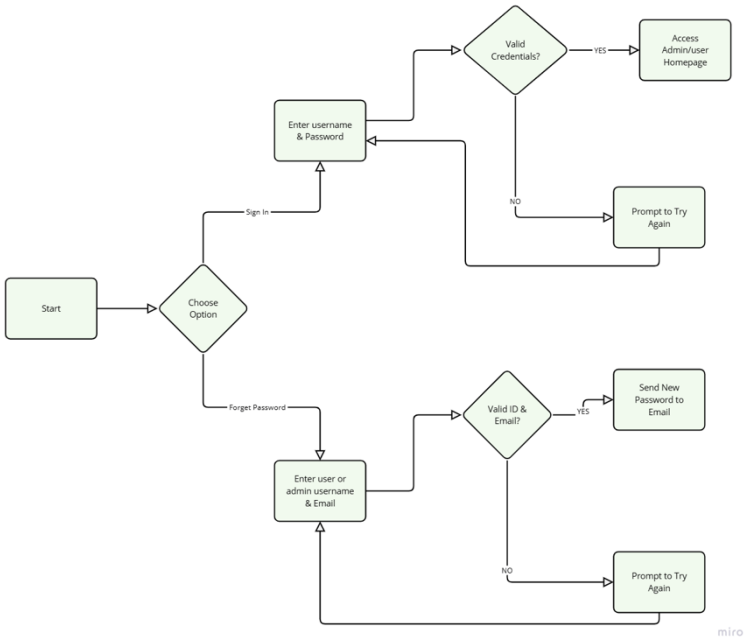


Figure 12: sign in and Reset password(forget password) flow chart

3.3.7 Use Case: Get Real-Time Recommendations

Attribute	Details
Actors	End User
Description	Provides real-time recommendations to users while they interact with the whiteboard, such as during drawing or collaborative activities.
Priority	High
Data	User input (e.g., drawings, text), context information
Stimulus	User performs actions on the whiteboard (e.g., drawing, writing).
Response	System generates and displays relevant recommendations in real-time based on user activity.
Abnormal Conditions	- If recommendations fail to load due to connectivity issues. - If user input is unclear or not recognized, resulting in irrelevant suggestions. - If the system encounters a processing error while generating recommendations.

Table 37 Use Case: Get Real-Time Recommendations

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3.3.8 Use Case: Use\interact with whiteboard

Attribute	Details
Actors	End User
Description	A collaborative whiteboard tool that allows users to draw, write, and interact in real time, with options for real-time recommendations and video generation based on user choices.
Priority	High
Data	User input (e.g., drawings, text), collaboration settings, video preferences
Stimulus	The user clicks on the "Interactive Whiteboard" button
Response	The whiteboard interface opens. - user can annotate or draw on the board. - Any drawings or text entered are saved in the session data. - System provides real-time recommendations during user interactions. - If chosen, the system can generate video recordings of the session. - Users can collaborate in real time with others.
Abnormal Conditions	- If real-time recommendations fail due to connectivity issues. - If video generation encounters errors (e.g., insufficient resources). - If user input is unclear, leading to inaccurate recommendations. - If collaboration settings cannot be applied due to permissions issues. - If the whiteboard fails to load, an error message is displayed, and the user is prompted to try again or contact support.

Table 38 Use Case: Use\interact with whiteboard

3.3.9 Use Case: Use Collaborative Tools

Attribute	Details
Actors	End User
Description	Enables users to collaborate in real time, sharing ideas and content .
Priority	High
Data	User input, shared documents, collaboration settings
Stimulus	User initiates collaboration through the whitboard.
Response	Users can interact, share, and edit content together in real time.
Abnormal Conditions	- If collaboration features fail to load due to connectivity issues. - If permissions are insufficient for certain users to collaborate.

Table 39 Use Case: Use Collaborative Tools

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3.3.10 Use Case: Use Video Generation

Attribute	Details
Actors	End User
Description	Allows users to generate video content from whiteboard sessions or text .
Priority	Medium
Data	User input (text)
Stimulus	User selects the option to generate a video.
Response	System creates and saves the video for user access.
Abnormal Conditions	- If video generation fails due to insufficient storage or error.

Table 40 Use Case: Use Video Generation

3.3.11 Use Case: Use Google Calendar

Attribute	Details
Actors	End User
Description	Integrates the platform with Google Calendar to manage schedules and events.
Priority	Medium
Data	User calendar events, reminders
Stimulus	User links their Google Calendar to the platform.
Response	Events from Google Calendar are displayed and can be managed within the platform.
Abnormal Conditions	- If the integration fails due to authentication issues. - If calendar data cannot be retrieved due to API errors.

Table 41 Use Case: Use Google Calendar

3.3.12 Use Case: Track Tasks

Attribute	Details
Actors	End User
Description	Allows users to create, assign, and monitor tasks within the platform.
Priority	High
Data	Task details, user assignments
Stimulus	User creates or updates a task.
Response	Tasks are tracked and displayed with progress updates.
Abnormal Conditions	- If task creation fails due to data validation errors. - If tasks cannot be retrieved due to system errors.

Table 42 Use Case: Track Tasks

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3.3.13 Use Case: Monitor Progress

Attribute	Details
Actors	End User
Description	Provides insights into user progress on tasks and learning objectives.
Priority	High
Data	User performance metrics, task completion rates
Stimulus	User requests progress reports.
Response	System generates and displays progress reports.
Abnormal Conditions	- If data cannot be retrieved due to system issues. - If metrics are inaccurate due to data processing errors.

Table 43 Use Case: Monitor Progress

3.3.14 Use Case: Set\Get Reminders

Attribute	Details
Actors	End User
Description	Sends reminders to users for tasks, events, and deadlines.
Priority	High
Data	Reminder details, user preferences
Stimulus	User sets a reminder for a specific task or event.
Response	Users receive reminders as scheduled.
Abnormal Conditions	- If reminders fail to send due to system errors - If user preferences are not saved correctly.

Table 44 Use Case: Set\Get Reminders

3.3.15 Use Case: Create Study Plans

Attribute	Details
Actors	End User
Description	Generates study plans based on user input and files, with the option to save them on the platform or export to Google Calendar.
Priority	High
Data	User preferences, file inputs, study objectives
Stimulus	User requests to generate a study plan.
Response	The system generates a study plan that can be saved to the platform or exported to Google Calendar based on user choice.
Abnormal Conditions	- If file formats are unsupported for study plan generation. - If there is an error in exporting to Google Calendar due to API issues. - If user input is insufficient to generate a plan.

Table 45 Use Case: Create Study Plans

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3.3.16 Use Case: Get Resource Recommendations

Attribute	Details
Actors	End User
Description	Provides users with recommendations for resources based on their learning needs.
Priority	Medium
Data	User preferences, resource database
Stimulus	User requests recommendations.
Response	System generates and displays tailored resource suggestions.
Abnormal Conditions	- If recommendations are not available due to data issues. - If user preferences are not recognized.

Table 46 Use Case: Get Resource Recommendations

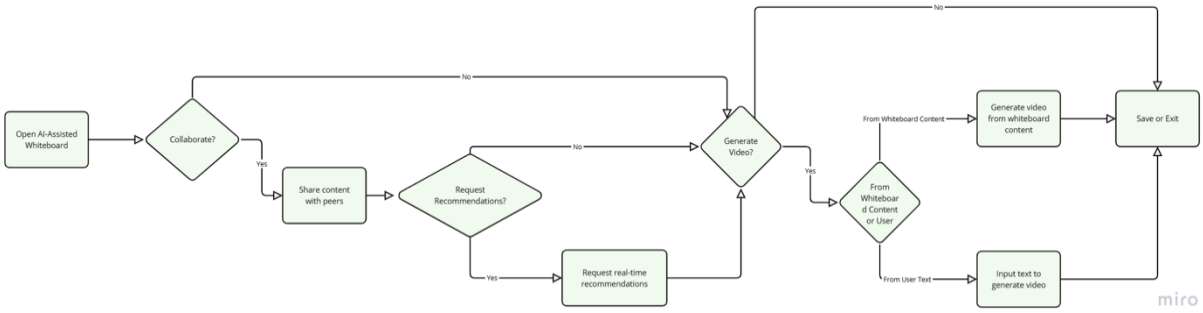


Figure 13 Use Cases flow chart

3.3.17 Use Case: Authorize User

Attribute	Details
Actors	End User , Admin
Description	Allows users to gain access to the platform by validating their credentials.
Priority	High
Data	Username, Password, Authentication Tokens
Stimulus	User attempts to log in or access a restricted area.
Response	System verifies credentials and grants or denies access based on the validation outcome.
Abnormal Conditions	- If invalid credentials are entered, an error message is displayed. - If the account is locked or inactive, an appropriate message is shown. - If the system is unable to connect to the authentication service, an error message is displayed.

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Table 47 Use Case: Authorize User

3.4 Classes / Objects

This section outlines the key classes and objects within the EDUassist. Each class represents a core component of the software, encapsulating relevant attributes and methods that facilitate the application's functionalities.

3.4.1 End User

Attributes:

- userID: Unique identifier for each end user.
- username: Name of the end user.
- password: Encrypted password for authentication.
- email: User's email address.
- phone: User's phone number.
- dateOfBirth: User's date of birth.
- studyPlans: List of study plans created by the user.
- resources: List of educational resources accessed by the user.

Functions:

- signIn(): Authenticates the user and grants access to the system.
- signOut(): Logs the user out of the system.
- resetPassword(): Initiates password recovery process.
- updateProfile(): Enables the user to modify their personal information.
- generateStudyPlan(): Allows the user to create a personalized study plan.
- manageStudyPlan(): Provides functionality to add, update, and track tasks within the study plan.
- accessResources(): Allows the user to access the library of educational resources.
- shareResources(): Enables the user to share resources with other users.
- collaborateOnWhiteboard(): Allows the user to utilize the AI-assisted whiteboard for visualization and collaboration.
- generateExplanationVideos(): Enables the user to create short videos from whiteboard
- trackProgress(): Monitors the user's progress on their study plans.

<Reference to functional requirements and/or use cases>

Functional Requirement: The system must provide end users with comprehensive functionality to manage their academic responsibilities, access educational resources, and collaborate effectively.

Use Case: An end user signs in to the EDUassist system, updates their profile information, generates a new study plan, accesses recommended resources, and collaborates with peers on the AI-assisted whiteboard.

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3.4.2 Admin

Attributes:

- adminID: Unique identifier for each admin.
- username: Admin's username.
- password: Encrypted password for admin authentication.
- email: Admin's email address.

Functions:

- signIn(): Allows the admin to authenticate and access the system.
- signOut(): Logs the admin out of the system.
- createUser(): Enables the admin to add new user accounts.
- updateUser(): Allows the admin to modify existing user profiles.
- deleteUser(): Enables the admin to remove user accounts from the system.
- resetUserPassword(): Allows the admin to reset a user's password.
- manageRoles(): Enables the admin to assign and update user roles (e.g., student).
- monitorSystemPerformance(): Provides the admin with tools to monitor system usage and stability.
- configureSystemSettings(): Allows the admin to customize system-wide settings and integrations.

<Reference to functional requirements and/or use cases>

Functional Requirement: The system must provide administrators with the necessary tools to manage user accounts, monitor system performance, and configure system-wide settings.

Use Case: An admin signs in to the EDUassist system, creates a new user account, assigns the user a specific role, and configures an integration with the institution's learning management system.

3.4.3 StudyPlan

Attributes:

- planID: Unique identifier for each study plan.
- userID: Identifier for the user associated with the plan.
- title: Title or name of the study plan.
- tasks: List of tasks and assignments included in the plan.
- deadlines: Deadlines associated with the tasks and assignments.
- progress: Tracking of the user's progress on the plan.

Functions:

- createPlan(): Allows the user to generate a new personalized study plan.
- updatePlan(): Enables the user to modify an existing study plan.
- addTask(): Lets the user add a new task or assignment to the plan.
- updateTask(): Allows the user to modify the details of a specific task.
- trackProgress(): Enables the user to monitor their progress on the study plan.
- getRecommendations(): Provides the user with personalized recommendations for resources based on the study plan.

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<Reference to functional requirements and/or use cases>
Functional Requirement: The system must support users in creating, managing, and tracking their personalized study plans, as well as provide relevant resource recommendations.
Use Case: A user creates a new study plan, adds tasks and deadlines, tracks their progress, and receives personalized recommendations for educational resources to support their academic goals.

3.4.4 Resource

Attributes:

- resourceID: Unique identifier for each educational resource.
- title: Title of the resource.
- type: Type of resource (e.g., article, video, tutorial, book).
- description: Brief description of the resource content.
- url: Link or file path to access the resource.
- tags: Keywords or tags associated with the resource.
- rating: User-based rating for the resource.

Functions:

- searchResources(): Allows users to search for relevant educational resources.
- viewResource(): Enables users to access and view the details of a specific resource.
- rateResource(): Allows users to provide a rating for a resource they have used.
- recommendResources(): Provides personalized resource recommendations based on the user's study plan and preferences.
- shareResource(): Enables users to share resources with their peers.

<Reference to functional requirements and/or use cases>
Functional Requirement: The system must provide users with access to a comprehensive library of educational resources and the ability to discover, rate, and share these resources.
Use Case: A user searches for resources on a specific topic, views the details of a recommended resource, provides a rating, and shares the resource with their study group.

3.4.5 Whiteboard

Attributes:

- whiteboardID: Unique identifier for each whiteboard session.
- userIDs: List of user IDs participating in the whiteboard session.
- content: Saved content and annotations from the whiteboard session.
- timestamp: Timestamp for the whiteboard session.
- recommendations: Real-time recommendations based on the session's content and user interactions.

Functions:

- createWhiteboard(): Allows users to initiate a new whiteboard session.
- joinWhiteboard(): Enables users to join an existing whiteboard session.

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- annotateWhiteboard(): Provides tools for users to draw, write, and annotate the whiteboard content.
- saveWhiteboard(): Saves the content and annotations from the whiteboard session.
- generateExplanationVideo(): Converts the whiteboard content, including text and annotations, into a short explanatory video designed to clarify complex concepts.
- provideRealTimeRecommendations(): Analyzes the content and interactions during the session to suggest relevant resources or concepts in real-time.

<Reference to functional requirements and/or use cases>

Functional Requirement: The system must include an AI-assisted whiteboard feature that allows users to collaborate, visualize concepts, and generate explanatory videos from text and annotations to aid in understanding complex topics.

Use Case: A group of students collaborates on the whiteboard to discuss a complex concept. They annotate key points and write explanations. They can also use the "generate explanation video" function to create a video that visually represents some concepts, making it easier to understand.

3.4.6 Task

Attributes:

- taskID: Unique identifier for each task.
- title: Title of the task.
- description: Detailed description of the task.
- deadline: Due date for the task.
- status: Current status of the task (e.g., pending, completed).
- priority: Priority level of the task (e.g., low, medium, high).
- userID: Identifier for the user associated with the task.

Functions:

- createTask(): Allows the user to create a new task.
- updateTask(): Enables the user to modify the details of an existing task.
- deleteTask(): Allows the user to remove a task from their study plan.
- markAsCompleted(): Updates the status of the task to completed.
- getTaskDetails(): Retrieves detailed information about a specific task.

<Reference to functional requirements and/or use cases>

Functional Requirement: The system must allow users to create, update, track, and manage their tasks effectively within their study plans.

Use Case: A user creates a new task, sets a deadline, marks it as completed after finishing it, and views the details of their remaining tasks.

3.4.7 Video

Attributes:

- videoID: Unique identifier for each video.
- title: Title of the video.
- description: Brief description of the video's content.

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- url: Link or file path to access the video.
- duration: Length of the video.
- createdBy: Identifier for the user who created the video.
- relatedTopics: List of topics related to the video's content.

Functions:

- uploadVideo(): Allows users to upload a new video to the system.
- editVideo(): Enables users to modify details of an existing video.
- deleteVideo(): Allows users to remove a video from the system.
- viewVideo(): Enables users to access and watch the video.
- generateVideoFromContent(): Creates a new video based on the content from the whiteboard or text annotations.

<Reference to functional requirements and/or use cases>

Functional Requirement: The system must allow users to create, manage, and view educational videos, enhancing the learning experience through visual content.

Use Case: A user uploads a video explaining a complex topic, edits the description for clarity, and shares it with their study group for review.

3.4.8 Recommendation

Attributes:

- recommendationID: Unique identifier for each recommendation.
- userID: Identifier for the user receiving the recommendation.
- contentType: Type of content being recommended (e.g., resource, video, study plan).
- contentID: Identifier for the recommended content.
- reason: Explanation for why the recommendation is made.
- timestamp: Time when the recommendation was generated.

Functions:

- generateRecommendations(): Analyzes user behavior and study plans to suggest relevant resources and videos.
- viewRecommendations(): Allows users to view a list of their recommendations.
- provideFeedback(): Enables users to give feedback on the usefulness of the recommendations.

<Reference to functional requirements and/or use cases>

Functional Requirement: The system must provide personalized recommendations to users based on their study habits and interests, enhancing their learning experience.

Use Case: A user receives a recommendation for a video resource based on their current study plan and past interactions, helping them to better understand a difficult topic.

3.4.9 Notification

Attributes:

- notificationID: Unique identifier for each notification.
- userID: Identifier for the user receiving the notification.
- message: Content of the notification.

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- type: Type of notification (e.g., reminder, update, alert).
- timestamp: Time when the notification was created.
- status: Status of the notification (e.g., read, unread).

Functions:

- createNotification(): Allows the system to generate a new notification for the user.
- retrieveNotifications(): Enables users to view their list of notifications.
- markAsRead(): Updates the status of a notification to read.
- deleteNotification(): Allows users to remove a notification from their list.

<Reference to functional requirements and/or use cases>

Functional Requirement: The system must provide users with notifications regarding important updates, reminders, and alerts related to their academic responsibilities.

Use Case: A user receives a notification reminding them of an upcoming deadline for a task in their study plan and checks their notifications list for any updates.

3.4.10 Progress

Attributes:

- progressID: Unique identifier for each progress record.
- userID: Identifier for the user whose progress is being tracked.
- studyPlanID: Identifier for the associated study plan.
- completedTasks: Number of tasks completed in the study plan.
- totalTasks: Total number of tasks in the study plan.
- progressPercentage: Percentage of completion for the study plan.
- lastUpdated: Timestamp for the last progress update.

Functions:

- trackProgress(): Allows users to track their progress on study plans and tasks.
- updateProgress(): Enables users to update their progress as they complete tasks.
- viewProgressReport(): Provides users with a detailed report of their progress over time.

<Reference to functional requirements and/or use cases>

Functional Requirement: The system must allow users to track their academic progress and provide insights into their completion rates for study tasks and plans.

Use Case: A user updates their progress after completing several tasks in their study plan, views their progress report, and identifies areas where they need to focus more effort.

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3.5 Non-Functional Requirements

The non-functional requirements define the system attributes that impact usability, reliability, and performance, ensuring EDUassist meets user expectations and operational standards. Each attribute is defined in measurable terms to guarantee that the system is robust and efficient.

3.5.1 Performance

The EDUassist system should handle user interactions efficiently, with 95% of all operations (e.g., loading study plans, accessing resources) completing within 2 seconds. System response time should not exceed 5 seconds even during peak usage hours.

3.5.2 Reliability

The system must maintain an uptime of at least 99.9% per month, ensuring that users have consistent access to study materials and tools. Data storage and backup processes should prevent any loss of user data, with automatic recovery options in case of failure.

3.5.3 Availability

EDUassist should be accessible 24/7, with scheduled maintenance limited to non-peak hours. In cases of downtime, users should receive advance notifications with expected availability times.

3.5.4 Security

The system will have secure login, data encryption, and user roles to keep data safe. It will also follow important security standards, like **GDPR** (General Data Protection Regulation), which protects users' personal information, All data exchanges will be secured with **SSL** (Secure Sockets Layer), a protocol that encrypts data to prevent unauthorized access.

3.5.5 Maintainability

The codebase should be modular and documented to allow for future updates and maintenance with minimal disruptions. Bug fixes and software patches should be applied within 48 hours of issue identification.

3.5.6 Portability

The EDUassist platform should support multiple operating systems, including iOS, Android, Windows, and macOS, ensuring cross-platform functionality. It should be optimized for both desktop and mobile devices, with responsive design principles for seamless use on various screen sizes.

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3.6 Inverse Requirements

Inverse requirements delineate the conditions under which the EDUassist Platform should refrain from certain actions or functionalities, thereby establishing critical boundaries and mitigating undesirable outcomes. The following inverse requirements have been identified:

1. **No Unauthorized Access:** The platform must prevent users from accessing features or data beyond their designated permissions, thereby ensuring that only authorized individuals can view or modify sensitive information.

2. **No Data Loss:** The platform must guarantee the integrity of user data during interactions, including processes related to sign-ins, content creation, and data storage, thereby safeguarding against any potential data loss.

3. **No Downtime During Peak Hours:** The platform should maintain operational continuity and avoid significant performance degradation during periods of high usage, ensuring consistent availability for all users.

4. **No Inaccurate Recommendations:** The platform must not generate irrelevant or inaccurate recommendations for study materials, thereby upholding the quality and relevance of suggested resources to enhance the user experience.

5. **No Overlapping Tasks:** The task tracking feature must prevent users from establishing overlapping deadlines for the same assignments, thus avoiding confusion and ensuring clarity in task management.

6. **No Unclear User Interface:** The platform must not present a confusing or unintuitive user interface that could impede the user experience, ensuring that navigation and functionality remain straightforward and accessible.

7. **No Incomplete Notifications:** The platform must avoid sending notifications that lack essential information, thereby ensuring that users receive comprehensive and actionable reminders regarding deadlines and study sessions.

By articulating these inverse requirements, the EDUassist Platform can effectively guide its development and operational strategies to circumvent common pitfalls, thereby enhancing user satisfaction and ensuring the overall integrity of the platform.

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3.7 Design Constraints

The design of EDUassist is influenced by several limitations and requirements that impact how the system is planned. These include:

1. **Documentation-Only Scope:**

This project focuses exclusively on the documentation and theoretical design of EDUassist. So no actual software implementation is included, which limits the ability to test and validate certain assumptions.

2. **Standards Compliance:**

EDUassist must follow privacy laws like General Data Protection Regulation (GDPR) to protect user data and meet accessibility standards

3. **Hardware and Software Compatibility:**

The system is designed to work on common devices like smartphones and laptops running modern operating systems where older devices might not support all features.

4. **Integration Requirements:**

The system depends on tools like Google Calendar and Learning Management Systems (LMS). Any changes to these tools, like updates to their APIs, could affect how EDUassist works with them.

5. **Team Resources and Expertise**

The team has limited resources and technical skills, so the design avoids advanced features that would need specialized knowledge, like complex AI models.

6. **Budget Constraints:**

As a theoretical project, the design assumes limited funding, Expensive infrastructure or tools are not included in the design.

7. **Scalability and Performance:**

The system is designed to handle a growing number of users, but since it’s not being implemented, these scalability assumptions can’t be tested.

8. **Security Measures:**

The design must incorporate strong security protocols, including multi-factor authentication and encrypted data storage, to protect user information. However, detailed implementation of these features is beyond the current scope.

9. **Language and Cultural Support:**

While the system is designed to theoretically support multiple languages, including Arabic, detailed customizations for specific cultural or regional needs is outside the project’s scope.

10. **Time Constraints:**

The project documentation must be completed within a predefined timeline, which limits how much we can explore or add to the design.

These constraints help shape the design of EDUassist, making sure it stays realistic and focused while still meeting user needs and following important standards.

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3.8 Logical Database Requirements

The logical database requirements outlined above ensure that the EDUassist Platform will effectively manage data while maintaining integrity, security, and compliance with user needs. By specifying formats, retention policies, and access controls, the platform establishes a robust foundation for its operations and user interactions. The EDUassist Platform will be designed to utilize a relational database to manage and store various types of data related to users, educational resources, interactions, and system functionalities without actual implementation at this stage.

3.8.1. Logical Requirements

1. **Data Formats**

   The following table outlines the data formats used in the database:

Data Type	Format/Specification	Description
User Data	UTF-8	Standard character encoding for text data.
Date and Time	ISO 8601 (YYYY-MM-DD)	Standard format for storing date and time.
Password	Hashed (e.g., bcrypt)	Secure storage of user passwords.
Email Address	Validated String	Ensures proper email format.
Resource URLs	String (VARCHAR)	Stores links to educational resources.
Video Content	Binary Large Object (BLOB)	Stores multimedia content securely.

Table 48 Data Formats

3.8.2. Storage Capabilities

- The database will support a scalable architecture to accommodate a growing number of users and resources.
- It must efficiently handle large volumes of data, especially multimedia resources like videos and images.
- Cloud-based storage options may be considered to enhance scalability and accessibility.

3.8.3. Data Retention

The following table summarizes the data retention policies:

Data Type	Retention Period	Description
User Activity Logs	Minimum of 1 year	For auditing and monitoring purposes.
User Profiles	Indefinite	Retained until user account deletion.
Educational Resources	Indefinite	Maintained for continuous access by all users.
User-Generated Content	Indefinite	Retained unless deleted by the user or admin.

Table 49 Data Retention

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3.8.4. Data Integrity

- The database will enforce referential integrity to maintain consistent relationships between tables (e.g., users and their study plans).
- Constraints such as unique constraints will be implemented on usernames and email addresses to prevent duplicate entries.
- The database must ensure that all transactions adhere to ACID properties (Atomicity, Consistency, Isolation, Durability) to protect against data corruption.

3.8.5. Backup and Recovery

- The database will implement regular automated backups to prevent data loss.
- A recovery plan will be in place to restore data quickly in the event of a system failure or data corruption.

3.8.6. Access Control

- The database will support role-based access control (RBAC) to ensure that only authorized users can perform operations on sensitive data.
- Audit logs will be maintained to track data access and modifications, ensuring accountability and security.

3.8.7. ERD and UML Class diagrams :

Entity	Attributes
User	UserID (PK), Username (UNIQUE), Password, Email (UNIQUE), Role, CreatedAt, UpdatedAt
Admin	AdminID (PK), Username, Password, Email, CreatedAt, UpdatedAt
StudyPlan	StudyPlanID (PK), UserID (FK), PlanDetails, CreatedAt, UpdatedAt
Task	TaskID (PK), UserID (FK), Title, Description, DueDate, Status, CreatedAt, UpdatedAt
Notification	NotificationID (PK), UserID (FK), Message, IsRead, CreatedAt
Progress	ProgressID (PK), UserID (FK), StudyPlanID (FK), Achievements, CreatedAt, UpdatedAt
Resource	ResourceID (PK), Title, URL, Type, CreatedAt, UpdatedAt
Whiteboard	WhiteboardID (PK), UserID (FK), StudyPlanID (FK), Content, CreatedAt, UpdatedAt
Video	VideoID (PK), UserID (FK), WhiteboardID (FK), VideoURL, CreatedAt, UpdatedAt
Recommendation	RecommendationID (PK), UserID (FK), ResourceID (FK), Reason, CreatedAt

Table 50 ERD and UML Class diagrams

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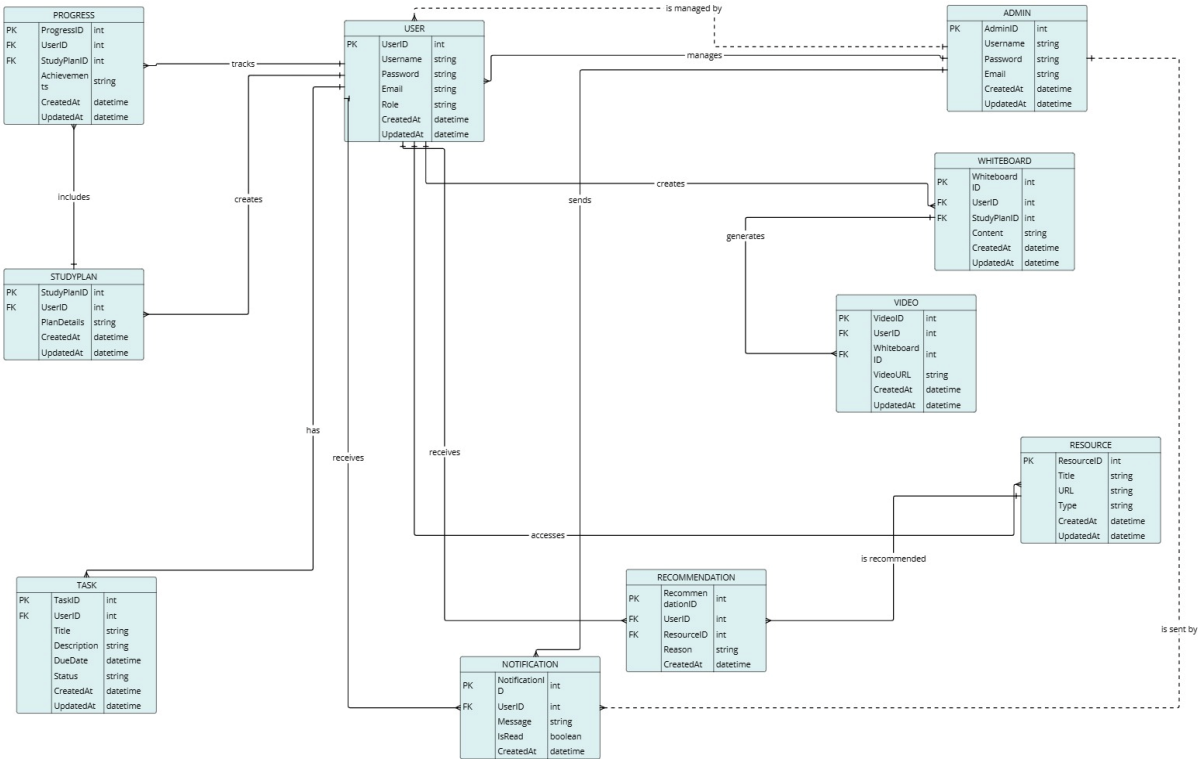


Figure 14 ERD for EDUassent

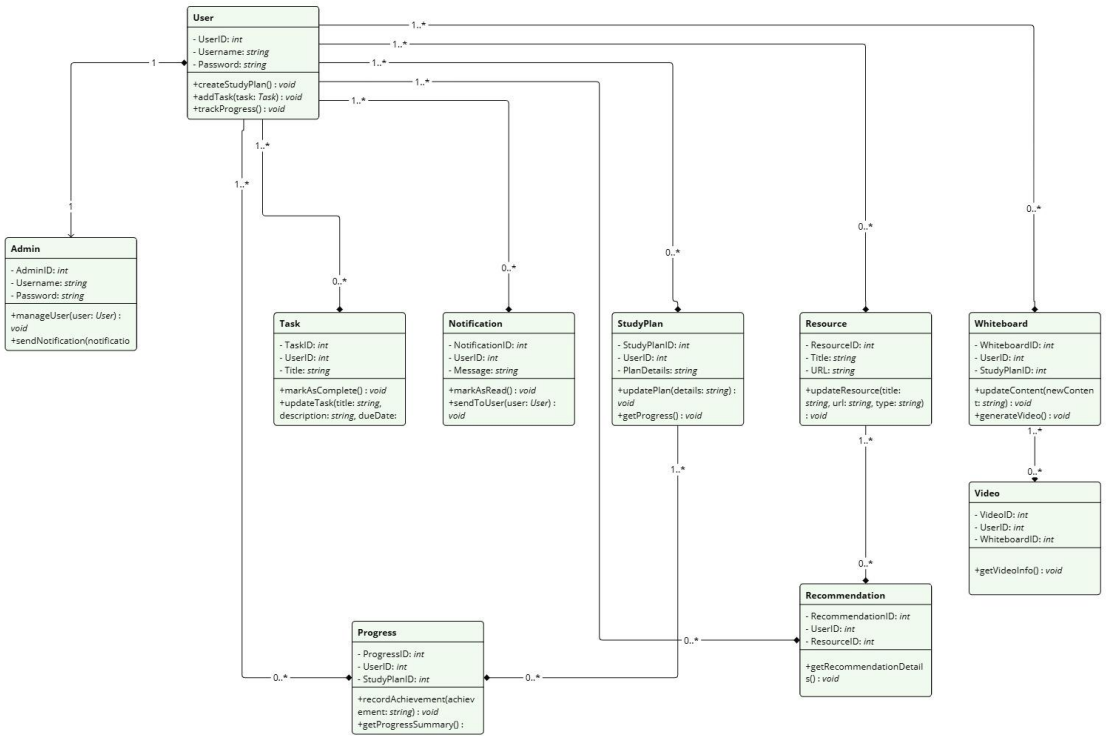


Figure 15 UML class diagram

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Relationship	Entity 1	Entity 2	Type	Description
One-to-Many	User	StudyPlan	1:N	A user can have multiple study plans.
One-to-Many	User	Task	1:N	A user can have multiple tasks.
One-to-Many	User	Progress	1:N	A user can have multiple progress records.
One-to-Many	StudyPlan	Progress	1:N	A study plan can have multiple progress records.
One-to-Many	User	Notification	1:N	A user can receive multiple notifications.
One-to-Many	User	Resource	1:N	A user can access multiple resources.
One-to-Many	User	Whiteboard	1:N	A user can create multiple whiteboards.
One-to-Many	Whiteboard	Video	1:N	A whiteboard can have multiple videos generated from its content.
One-to-Many	User	Recommendation	1:N	A user can receive multiple recommendations.
One-to-Many	Resource	Recommendation	1:N	A resource can be recommended to multiple users.
One-to-Many	Admin	User	1:N	An admin can manage multiple user accounts.
One-to-Many	Admin	Notification	1:N	An admin can send notifications to multiple users.

Table 51 Entities Relationships

The Entity-Relationship Diagram (ERD) for the EDUassist provides a structured overview of the data model, highlighting key entities, their attributes, and the relationships that govern the interactions within the system. This detailed representation serves as a guide for database design, ensuring that all necessary data points and relationships are clearly defined .

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4. Analysis Models

4.1 Sequence Diagrams

4.1.1. EDUassent main functionates

The diagram outlines the main features and user interactions, assuming the precondition of an active user account on the platform.

The core functionalities highlighted in the diagram include:

1.Personalized recommendations and videos for the user.

2.The ability to request study recommendations.

3.Tailored study suggestions based on the user's syllabus.

4.The use of an interactive whiteboard.

5.Provision of real-time study suggestions.

6.Compilation and sharing of whiteboard content, including explanatory videos.

7.Requests for resource suggestions.

8.Notifications, reminders for deadlines, and the ability to suggest alternative topics.

The diagram also indicates that the system will handle error scenarios, displaying an error message if the platform is temporarily unavailable. Overall, the sequence diagram provides a comprehensive visual representation of the key user interactions and system functionalities within the EDUassist educational platform, showcasing its ability to offer personalized learning experiences and a range of interactive tools to support userss' educational needs.

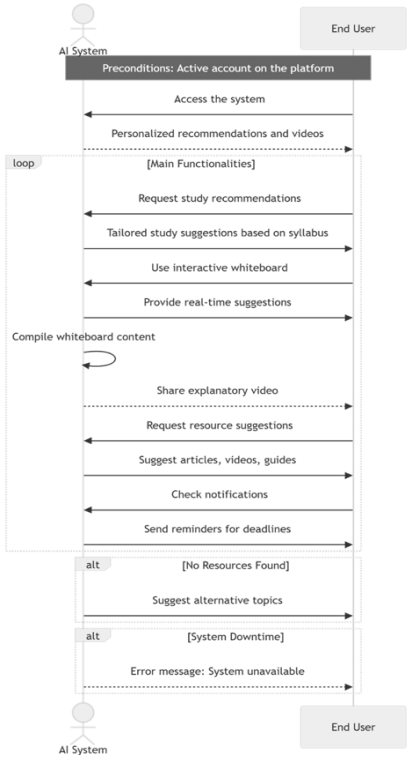


Figure 16EDUassist main functionates

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4.1.2. Flows for EDUassist Components

1. **Interactive Whiteboard:** The diagram illustrates interactions with the interactive whiteboard, enabling users to access, annotate, and visualize concepts while receiving real-time suggestions. This feature fosters engagement and supports collaborative learning.
2. **AI Video Generator:** This component allows users to request educational video creation. The system compiles the content and shares the final product. If an error occurs, a relevant message is displayed to the user.
3. **Study Plan Generator:** This feature enables users to input syllabi, which the system processes to create a customized study plan. Users can edit the plan, receiving feedback on the generation process's success or failure.
4. **Authentication and Configuration:** The diagram represents workflows for user authentication and system settings. Users and administrators log into the platform, manage preferences, and update features, with the system providing confirmations or error notifications as necessary.
5. **Comprehensive Functionality:** Overall, the sequence diagram captures the coordinated interactions among various components of the EDUassist platform, highlighting its capabilities in collaborative tools, content generation, personalized planning, and administrative management, all essential for enhancing educational experiences.

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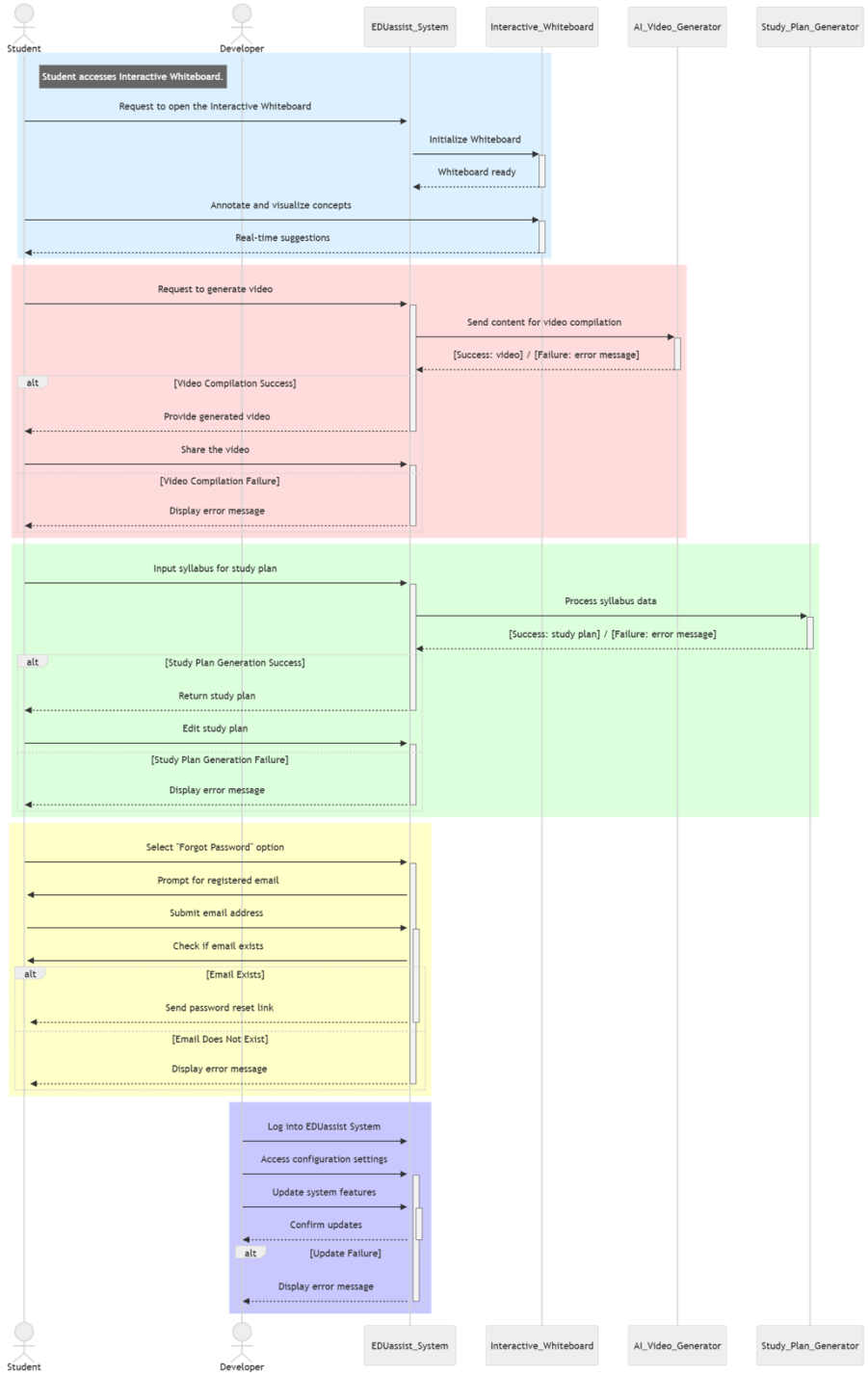


Figure 17 EDUassist main flow

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4.2 STATE-TRANSITION DIAGRAMS (STD)

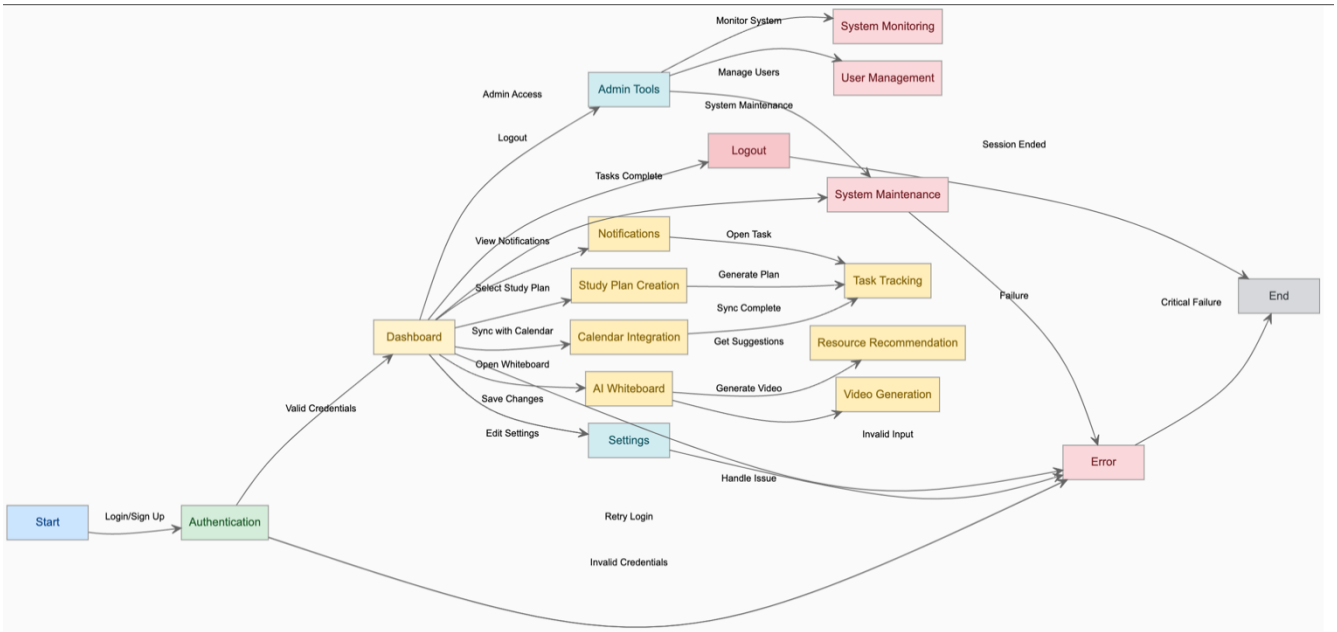


Figure 18 state transition diagram (STD)

4.3 Data Flow Diagrams (DFD)

The Data Flow Diagram (DFD) for EDUassist illustrates how data flows through the system, focusing on interactions between Students, Admins, and Third-Party Tools like Google Calendar and Zoom. It breaks the system into key processes, such as creating user profiles, generating and updating study plans, monitoring progress, providing AI-driven recommendations, collaboration, and notifications.

Students input data like syllabus files and preferences, which are processed into personalized study plans and stored in the system. Admins oversee platform management and ensure smooth operations. The system also integrates with third-party tools for features like syncing schedules and collaboration. This DFD simplifies the system's structure and highlights how data moves to support personalized learning and efficient task management.

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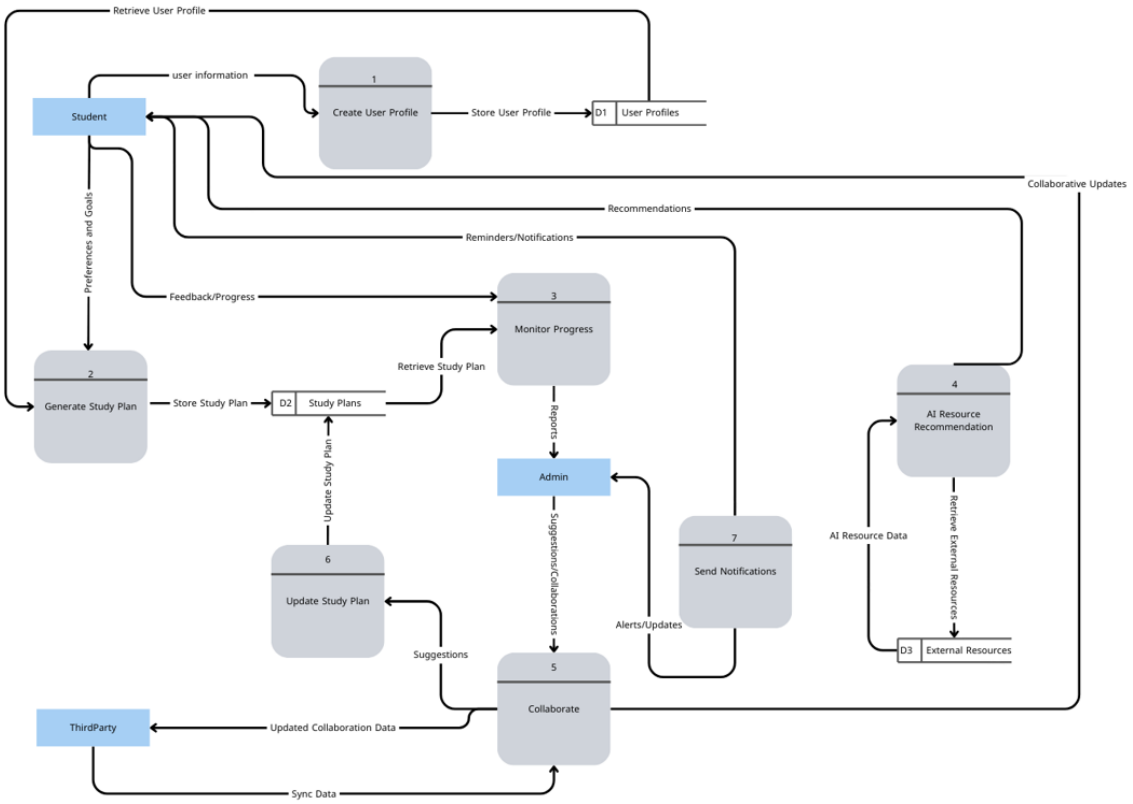


Figure 19Data Flow Diagrams (DFD)

5. Change Management Process

The Change Management Process for the EDUassent SRS outlines the steps to be taken when there are changes to the project scope or requirements. This process ensures that all changes are systematically evaluated, documented, and approved to maintain the integrity of the EDUassent project and its requirements.



Figure 20 Requirements Change Management

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5.1 Change Submission

1. Who Can Submit Changes:
    - Project Managers
    - Development Team Members
    - Quality Assurance Specialists
2. Means of Submission:
    - Changes must be submitted through a formal change request form
    - The form should include a detailed description of the proposed change, the rationale behind it, and an assessment of the potential impact

5.2 Change Approval Process

1. Initial Review:
    - The Change Control Board (CCB) will review the change request
    - The CCB will typically include representatives from the project management team, faculty, instructional design, and educational technology
2. Impact Analysis:
    - The CCB will conduct an impact analysis to assess the technical feasibility of the change, the impact on course content, delivery, and student learning outcomes
    - The analysis will also consider the effect on project scope, timeline, and resources
3. Decision Making:
    - Based on the impact analysis, the CCB will decide to approve the change, reject the change, or request further information
    - All decisions will be documented in the project management system
4. Documentation:
    - The original change request, the outcome of the review, and any new or revised requirements will be added to the SRS

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- Version control will be used to track changes to the SRS

5. Notification:
    - Stakeholders will be notified of the decision regarding the change request
    - If approved, the updated SRS will be distributed to all relevant parties

5.3 Implementation of Changes

1. Updating the SRS:
    - Approved changes will be incorporated into the SRS, following established formatting and documentation guidelines
2. Version Control:
    - Each update to the SRS will be tracked using version control
    - The version history will include the version number, a summary of changes, the date, and the individuals involved
3. Review and Validation:
    - The updated SRS will undergo a validation process to ensure the changes accurately reflect the stakeholders' needs and the project objectives

5.4 Continuous Improvement

- The Change Management Process will be reviewed periodically to identify opportunities for improvement
- Feedback from stakeholders will be solicited to ensure the process remains efficient and effective in managing changes to the EDUassent SRS

By following this structured change management process, the EDUassent project team can adapt to evolving academic requirements while maintaining clarity and control over the project scope, ensuring successful delivery and stakeholder satisfaction.

A. Appendices

This section provides detailed information on the AI models and technologies that will be implemented within the **EDUassist** platform. Each model and technology is selected to support specific functionalities in the system, including personalized recommendations, video generation,

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and task management. The following descriptions offer an overview of each model's purpose and functionality to enhance user experience and optimize learning outcomes.

A.1 AI Models and Technologies

This appendix outlines the AI models and technologies utilized in **EDUassist** to create a responsive, intuitive, and efficient educational platform. These technologies enable real-time assistance, multimedia content creation, and effective task management, supporting a personalized and engaging learning experience.

A.1.1 Natural Language Processing (NLP) Models

1. **GPT (Generative Pre-trained Transformer)**
    - **Purpose:** To generate text-based recommendations and respond to user inquiries in real-time.
    - **Functionality:** GPT analyzes user inputs, understands context, and generates relevant educational content or suggestions, such as answering questions, summarizing text, or providing personalized study tips.
2. **BERT (Bidirectional Encoder Representations from Transformers)**
    - **Purpose:** To interpret the nuances in user queries and enhance recommendation accuracy.
    - **Functionality:** BERT is designed to handle context-aware responses, improving the system's ability to understand complex queries and deliver precise recommendations that align with the user's needs.

A.1.2 Video Generation Models

1. **OpenAI's DALL-E**
    - **Purpose:** To create educational videos or visual content from text-based descriptions.
    - **Functionality:** DALL-E translates written content or key concepts into visual formats, such as images or short animations, which enhances engagement and helps users visualize and better understand complex topics.

A.1.3 Recommendation Systems

1. **Collaborative Filtering**

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- **Purpose:** To provide personalized resource and task recommendations based on the behavior and preferences of similar users.
- **Functionality:** By analyzing patterns in user interactions across the platform, collaborative filtering identifies popular resources and activities among peers, delivering suggestions that may benefit the individual user.

2. **Content-Based Filtering**

- **Purpose:** To tailor recommendations by analyzing the characteristics of resources the user has previously engaged with.
- **Functionality:** This model customizes suggestions based on a user's unique history and preferences, offering similar or related content that aligns with their learning style and topics of interest.

A.1.4 Task Management and Sorting

1. **Machine Learning Algorithms**

- **Purpose:** To prioritize and organize tasks in the user's calendar based on deadlines, personal preferences, and historical study patterns.
- **Functionality:** Algorithms such as decision trees and support vector machines (SVMs) are used to dynamically classify and prioritize tasks. This ensures that users stay on track with assignments and study sessions by receiving reminders and seeing tasks sorted according to urgency and importance.

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